

VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)

Department of Computer Engineering



Project Report on Natalis - Prenatal Image Interpretation System

**In partial fulfilment of the Fourth Year, Bachelor of Engineering (B.E.)
Degree in Computer Engineering at the University of Mumbai
Academic Year 2025 - 26**

By

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(A.Y. 2025 - 26)

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Certificate

This is to certify that **Ved Waje(D17A - 64), Abhirat More (D17A - 44), Pranita Bannore (D17A - 8), & Harshita Lohana (D17A - 38)** of Fourth Year Computer Engineering studying under the University of Mumbai have satisfactorily completed the project on “**Natalis - Prenatal Image Interpretation System**” as a part of their coursework of PROJECT - II for Semester - VIII under the guidance of their mentor **Mrs. Vidya Zope** in the year 2025 - 26.

This thesis/dissertation/project report entitled “**Natalis - Prenatal Image Interpretation System**” by **Ved Waje, Abhirat More, Pranita Bannore, Harshita Lohana** is approved for the degree of **Bachelor of Engineering in Computer Engineering**.

Programme Outcomes	Grade
PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12 PSO1 & PSO2	

Date: 27th April, 2026

Project Guide: **Mrs. Vidya Zope**

Project Report Approval

For

B. E (Computer Engineering)

This project report entitled “**Natalis - Prenatal Image Interpretation System**” by **Ved Waje(D17A - 64), Abhirat More (D17A - 44), Pranita Bannore(D17A - 8), & Harshita Lohana(D17A - 38)** is approved for the degree of **Bachelor of Engineering** in **Computer Engineering**.

Examiners

1.

(Internal Examiner name & sign)

2.

(External Examiner name & sign)

3.

(Head of Department)

4.

(Principal)

Date: 27th April, 2026

Place: Chembur, Mumbai

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Date: 27th April, 2026

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We are thankful to our college Vivekanand Education Society's Institute of Technology for considering our project and extending help at all stages needed during our work of collecting information regarding the project.

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We wish to express our profound thanks to all those who helped us in gathering information about the project. Our families too have provided moral support and encouragement several times.

COURSE OUTCOMES FOR B.E. PROJECT

Learners will be able to,

Course Outcome	Description of the Course Outcome
CO1	Able to apply the relevant engineering concepts, knowledge and skills towards the project.
CO2	Able to identify, formulate and interpret the various relevant research papers and to determine the problem.
CO3	Able to apply the engineering concepts towards designing solutions for the problem.
CO4	Able to interpret the data and datasets to be utilized.
CO5	Able to create, select and apply appropriate technologies, techniques, resources and tools for the project.
CO6	Able to apply ethical, professional policies and principles towards societal, environmental, safety and cultural benefit.
CO7	Able to function effectively as an individual, and as a member of a team, allocating roles with clear lines of responsibility and accountability.
CO8	Able to write effective reports, design documents and make effective presentations.
CO9	Able to apply engineering and management principles to the project as a team member.
CO10	Able to apply the project domain knowledge to sharpen one's competency.
CO11	Able to develop a professional, presentational, balanced and structured approach towards project development.
CO12	Able to adopt skills, languages, environment and platforms for creating innovative solutions for the project.

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Abstract

Maternal mortality remains high in low-income regions ($\approx 295,000$ deaths in 2017) due to undiagnosed complications. The WHO recommends at least one ultrasound scan before 24 weeks' gestation, but up to half of pregnant women in resource-poor areas lack any screening. We propose Natalis, a smartphone-based AI ultrasound tool that analyzes only images and video to assist minimally trained health workers. Using portable ultrasound probes and deep learning, Natalis will guide the operator through a standardized scan and automatically compute gestational age, detect fetal presentation, identify placental location, count fetuses, and flag high-risk signs (e.g., placental previa, growth restriction) – similar to systems like BabyChecker and DeepEcho. Natalis is designed to run offline on an Android device with minimal training (e.g., a 3-min tutorial) and no constant power. We detail the full project plan – from data-driven model training and system architecture to hardware/software requirements and evaluation metrics (accuracy against experts, diagnostic sensitivity, etc.) – to demonstrate Natalis's potential as a deployable antenatal screening solution in low-resource settings.

Industry Letter



CARENX INNOVATIONS

Industry Letter

To
The HOD of the Computer Engineering Department
Vivekanand Education Society's Institute of Technology (VESIT),
Chembur, Mumbai

Subject: Submission of Major Project Details – “Natalis” (Ongoing) by CareNX Team

Respected Madam,

This letter serves as formal communication from CareNX, an organization committed to innovation in healthcare solutions. We wish to inform you about the major project titled “Natalis,” which is currently being undertaken by a talented group of students from your esteemed institution.

Project Title: Natalis
Project Status: Ongoing
Organization: CareNX
Project Team:

- Ved Waje
- Harshita Lohana
- Abhirat More
- Pranita Bannore

Natalis represents our joint efforts in advancing maternal and neonatal health through technical research and application. The team from VESIT is actively working under our guidance on this initiative, aimed at offering impactful solutions within the healthcare industry.

CareNX acknowledges the dedication and commitment shown by Ved Waje, Harshita Lohana, Abhirat More, and Pranita Bannore throughout the course of this project. We value the collaboration between CareNX and VESIT, and we look forward to continued support as the project progresses.

Thank you for your co-operation.
Warm regards,
CareNX avinash@carenx.com



Date: 13 October 2025

CARENX INNOVATIONS PRIVATE LIMITED

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Introduction

1.1 Introduction

Maternal health remains one of the most critical global health challenges, especially in low-income regions where access to quality prenatal care is limited. The World Health Organization (WHO) recommends that every pregnant woman receive at least one ultrasound scan before 24 weeks of gestation. Such screening is essential for accurately determining **gestational age (GA)**—a key indicator for monitoring fetal growth and identifying high-risk pregnancies. Accurate estimation of GA allows timely interventions and significantly reduces preventable maternal and neonatal complications.

Natalis is a proposed smartphone-based AI-assisted ultrasound system designed to estimate **gestational age** from standard ultrasound images or videos. The system empowers minimally trained health workers to perform basic obstetric screening using affordable, portable ultrasound probes. By analyzing only visual data, Natalis automates GA measurement and provides clear, interpretable results directly on a smartphone. Designed to function offline with minimal power requirements, Natalis aims to deliver an accessible, low-cost, and reliable prenatal screening solution for underserved and resource-limited settings.

1.2 Motivation for the Project

Despite significant medical advancements, approximately **295,000 women** die annually from preventable pregnancy-related complications, with **99% of these deaths** occurring in low- and middle-income countries. A major contributing factor is the lack of access to timely ultrasound screening, primarily due to the scarcity of trained sonographers and functional imaging equipment in rural clinics.

Recent innovations in AI and portable ultrasound technology demonstrate that diagnostic imaging can be democratized. Systems like **BabyChecker** and **DeepEcho** have proven that artificial intelligence can guide untrained health workers to acquire diagnostic-quality scans and interpret them accurately. Motivated by these breakthroughs, Natalis seeks to extend such capabilities to a wider audience through an intuitive Android application capable of real-time, on-device image analysis—empowering frontline workers and improving maternal health outcomes globally.

1.3 Drawbacks of the Existing System

Existing obstetric ultrasound systems face several limitations in low-resource settings:

- **Dependence on Expert Operators:** Traditional systems require trained sonographers, limiting use in remote areas.
- **High Equipment Costs:** Conventional ultrasound machines are expensive and require regular maintenance.

- **Power and Connectivity Constraints:** Many rural clinics lack reliable electricity or internet, making standard equipment unsuitable.
- **Complex Interfaces:** Most ultrasound systems are designed for clinical specialists, not community health workers.
- **Limited AI Integration:** Current low-cost portable devices often lack automated interpretation or guidance features, making them difficult to use effectively without expertise.

These constraints leave a large portion of the global population without access to essential prenatal screening.

1.4 Problem Definition

The key problem addressed by this project is the **lack of accessible, automated, and low-cost prenatal ultrasound screening** in resource-poor regions. Many expectant mothers never receive an ultrasound due to equipment unavailability or insufficient operator training.

Problem Statement:

To develop a **smartphone-based AI tool—Natalis**—capable of analyzing ultrasound images and videos to automatically estimate gestational age, detect fetal presentation, identify placental location, and count fetuses. The system must operate **offline**, require **minimal user training**, and function efficiently on low-power Android devices, thus enabling early risk detection and referral in underserved communities.

1.5 Relevance of the Project

The Natalis project is highly relevant to the goals of **global maternal health, AI in healthcare, and digital health equity**. By leveraging deep learning to automate image interpretation, Natalis can serve as a **scalable solution** for antenatal screening in rural and remote areas.

Its relevance extends to:

- **Public Health:** Contributes to the reduction of maternal and neonatal mortality.
- **Technology for Social Impact:** Demonstrates the practical use of AI in addressing real-world healthcare disparities.
- **Sustainability:** Operates offline with minimal infrastructure, aligning with the WHO's global digital health strategies.
- **Research Advancement:** Provides a data-driven framework for further exploration of AI-assisted ultrasound diagnostics.

Thus, Natalis bridges the gap between advanced medical imaging and community-level healthcare delivery.

1.6 Methodology Used

The project employs a data-driven deep learning methodology for ultrasound image analysis, combined with a mobile deployment framework for real-world usability. The development process includes the following stages:

1. **Data Collection and Preprocessing:** Ultrasound images and videos are sourced from public datasets and clinical collaborations. Data is preprocessed through denoising, normalization, and augmentation to improve model generalization.
2. **Model Development:** Convolutional Neural Networks (CNNs) and segmentation architectures (such as U-Net or DeepLab) are trained to perform multiple tasks—fetal biometric measurement, presentation classification, placental segmentation, and multi-fetus detection.
3. **System Integration:** The trained model is converted for on-device execution using frameworks like TensorFlow Lite or PyTorch Mobile. The model is integrated into the Natalis Android app, which features a simple user interface designed for health workers.
4. **Offline Inference:** The system performs AI-based analysis directly on the device without internet access, ensuring privacy and functionality in rural environments.
5. **Evaluation and Validation:** The AI outputs are compared against expert sonographer annotations using metrics such as accuracy, Dice coefficient (for segmentation), and mean absolute error (for gestational age estimation).
6. **Deployment and Testing:** Field testing is conducted to evaluate usability, robustness, and real-world diagnostic performance under varying environmental and hardware conditions.

Literature Survey

2.1 Research Papers

Table 1. Literature Survey of Research Papers

Sr No	Title of Paper	Journal/ Conference Date	Abstract	Inference Drawn
1	Estimating Fetal Gestational Age from Ultrasound Images Using Deep Learning Techniques	January 20–22, 2025	This study introduces the UI-DLT framework, a CNN-based system for automatically estimating fetal gestational age (GA) from ultrasound images. It reduces human error, operator dependency, and variability due to image quality or fetal position, achieving over 93% accuracy and 98% success rate.	Deep learning, specifically CNNs, can accurately and consistently estimate gestational age from ultrasound images, minimizing human dependency and enhancing reliability across clinical settings. The approach supports automation and scalability for low-resource healthcare environments.
	Hierarchical Attentive Network for Gestational Age Estimation in Low-Resource Settings	May 15-17, 2023	This study proposes a hierarchical deep sequence learning model with an attention mechanism to estimate gestational age (GA) using low-cost, handheld Doppler ultrasound signals. Data from 226 pregnancies in Guatemala were used, achieving an average error of only 0.79 months, near the theoretical minimum. The model highlights key fetal cardiac activity patterns to identify developmental stages and potential fetal growth restriction (FGR).	The attention-based hierarchical model effectively learns fetal cardiac dynamics for GA estimation using affordable Doppler devices. It shows strong potential for accurate, low-cost, and scalable GA estimation in low-resource settings, minimizing dependence on image quality or operator expertise.

Employing Support Vector Machine Regression to Estimate the Fetal Gestational Age	September 4–7, 2022	This study applies Support Vector Machine (SVM) regression using maternal and fetal heart rate variability (HRV) features extracted from 1-minute ECG recordings of 60 pregnant women. The linear kernel outperformed the RBF kernel, achieving an error of 5.11 weeks and a Pearson correlation of 0.63 with gold- standard GA (CRL). Results suggest maternal HRV plays a vital role in fetal age estimation.	Combining maternal and fetal HRV features in an SVM- linear model enhances GA estimation accuracy, emphasizing maternal cardiac influence on fetal development. Though promising, larger datasets and diverse conditions are needed to improve robustness and generalizability.
Fetal Head and Femur Detection from USG Image to Estimate Gestational Age	October 29–30, 2018	This study proposes an automated approach for detecting and segmenting fetal head and femur in ultrasound (USG) images using Integral Projection. Biparietal Diameter (BPD) and Femur Length (FL) were extracted as key parameters for gestational age (GA) estimation. The method achieved a 97.77% accuracy rate, improving consistency over manual observation.	Preprocessing reduced image noise, and segmentation using Integral Projection accurately identified fetal head and femur. Using BPD and FL as features provided high GA estimation accuracy. Future improvements could include additional parameters like abdominal circumference for enhanced precision.

Automatic Gestational Age Estimation by Femur Length Using Integral Projection from Fetal Ultrasonography	2018	This study presents an automated system for gestational age (GA) estimation using femur length extracted from fetal ultrasound images. Techniques like Gaussian Filtering, Morphological Operations, and Canny Edge Detection were applied for noise reduction and contour detection. Femur length was	The proposed automated method shows strong correlation with manual doctor measurements, minimizing human error and improving diagnostic reliability. It effectively measures femur length and estimates GA with high accuracy, demonstrating potential for clinical use in reducing observer variability.
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2.2. Software referred / Articles referred:

Baby Checker

Abstract

Baby Checker is an AI-powered smartphone-based ultrasound solution designed to address the critical global challenge of maternal mortality, particularly in low-resource settings. The system combines artificial intelligence with handheld ultrasound technology to enable frontline healthcare workers to perform pregnancy risk screening with minimal training. BabyChecker analyzes obstetric ultrasound scans to identify risks in pregnancy through a standardized six-sweep protocol across the abdomen, providing essential information including gestational age, fetal presentation, multiple gestations, and placenta location within 2 minutes.

The solution addresses the fact that approximately 800 women die daily from preventable pregnancy-related complications, with 99% of maternal deaths occurring in low-income countries where access to trained sonographers and ultrasound equipment is severely limited. BabyChecker operates offline on Android devices and requires only a 3-minute training video for healthcare workers to become proficient, making it accessible in areas with limited infrastructure and electricity.

Inference

Baby Checker has demonstrated significant clinical impact across multiple deployment sites. Field data shows that 76.3% of first-time users can capture usable scans immediately, indicating high usability from the initial attempt. The system has been successfully implemented in over twelve countries including Sierra Leone, Zambia, Honduras, Kenya, Malawi, and Tanzania, with validation studies confirming its effectiveness in real-world clinical settings.

The AI algorithms powering Baby Checker have been validated to perform pregnancy risk screening that enables early identification and management of high-risk pregnancies at local healthcare facilities. This technology facilitates task-shifting toward frontline health workers, empowering them to screen risky pregnancies more effectively than traditional approaches. The system's deterministic nature ensures consistent outputs, unlike human operators who may have variable performance.

Baby Checker functions as what has been described as a "great equalizer" by the Gates Foundation, enabling any frontline healthcare worker with no prior ultrasound experience to perform essential pregnancy screenings after minimal training. The solution has been recognized for its potential to bridge healthcare disparities and improve maternal health outcomes globally, particularly in underserved regions where traditional ultrasound services are unavailable.

Deep Echo

Abstract

DeepEcho is a cloud-based, standalone software as a medical device (SaMD) that utilizes deep learning algorithms to automate fetal biometry and amniotic fluid volume (AFV) assessments in fetal ultrasound screening. The system is designed to assist qualified healthcare professionals in the analysis and interpretation of fetal ultrasound images for patients aged 18 years or older between the 14th and 41st weeks of pregnancy. Deep Echo processes both DICOM images and live ultrasound streaming to identify standard biometry views (cephalic, abdominal, and femoral) and automatically compute critical measurements, including head circumference (HC), biparietal diameter (BPD), abdominal circumference (AC), femur length (FL), gestational age (GA), and estimated fetal weight (EFW).

The development of Deep Echo was motivated by the need to democratize access to quality fetal ultrasound, particularly addressing the high maternal and fetal mortality rates in regions with limited access to skilled sonographers. The system was trained using one of the world's largest fetal ultrasound annotated datasets, consisting of 172,293 de-identified Moroccan fetal ultrasound images supplemented with publicly available datasets.

Inference

Deep Echo has achieved strong clinical validation through rigorous prospective testing. In a landmark Nature Communications study, its 95% limits of agreement with experts were narrower than intra- and inter-observer variability among human sonographers across all parameters: $\pm 0.54\%$ (HC), $\pm 3.74\%$ (BPD), $\pm 0.14\%$ (AC), $\pm 3.11\%$ (FL), $\pm 1.45\%$ (GA), $\pm 2.42\%$ (EFW), and $\pm 16.96\%$ (SDP). Reliability was excellent, with ICC > 0.9 for all metrics except SDP (ICC = 0.692).

It outperformed state-of-the-art models like FUVAI in mean absolute error for most parameters, and achieved 86.6% sensitivity and 85.7% specificity for polyhydramnios detection. FDA 510(k) clearance in 2025 confirmed its safety and effectiveness. The deterministic design ensures consistent outputs, minimizing variability seen in human measurements.

Fetal ultrasound can lower fetal mortality by up to 20% and maternal mortality by 6%, underscoring the value of democratizing access. Deep Echo's automation enables minimally trained workers to deliver accurate prenatal assessments in resource-limited areas, addressing skilled sonographer shortages and expanding maternal healthcare access.

Requirements of the Proposed System

3.1. Functional Requirements:

1. User Interaction and Input

- o The system shall allow users to upload **pre-recorded ultrasound images or short videos** from local storage.
- o The system shall provide an **intuitive Android interface** suitable for non-technical users like midwives or field health workers.

2. AI-Based Image Analysis

- o The system shall perform **frame extraction** from uploaded videos.
- o The system shall identify **standard fetal planes** (head, abdomen, femur) using a **CNN classifier (ResNet)**.
- o The system shall **segment fetal structures** to measure biometric parameters (HC, AC, FL) using a **U-Net**.
- o The system shall detect **fetal orientation and count** (cephalic/breech, single/twins) using **YOLOv5**.
- o The system shall segment the **placental region** and identify its location (anterior/posterior/low-lying).

3. Report Generation

- o The system shall compute **gestational age (GA)** based on biometric parameters.
- o The system shall display results including:
 - Gestational Age
 - Fetal Presentation
 - Placental Location
 - Fetus Count
- o The system shall generate **color-coded risk alerts** and save reports for local record-keeping.

4. Offline Operation

- o The system shall function **entirely offline** without internet connectivity.
- o The AI inference shall run locally on the smartphone using **TensorFlow Lite / PyTorch Mobile**.

5. Usability & Accessibility

- o The interface shall support **multiple languages** and **icon-based visualization** for easy understanding.
- o The system shall provide a **quick setup (<3 minutes)** for non-expert health workers.

3.2. Non-Functional Requirements:

1. Performance

- o On-device inference shall complete within **2 minutes per scan**.
- o The app shall maintain **at least 20 FPS** during video frame extraction.

2. Accuracy

- o **GA estimation error $\leq \pm 2$ weeks**
- o **Fetal presentation detection >95% accuracy**
- o **Placenta segmentation Dice score ≥ 0.90**

3. Reliability

- o The app shall handle corrupted or low-quality images gracefully with warning messages.
- o The system shall maintain **data consistency** even if the app is interrupted mid-inference.

4. Scalability

- o The modular model architecture shall allow **easy integration** of future AI modules (e.g., IUGR, preeclampsia detection).

5. Usability

- o Designed for **minimal training requirements** and intuitive navigation.
- o Interface shall be **touch-optimized** for Android screens ≥ 5 inches.

6. Security

- o All user data (images, reports) shall remain **stored locally** to maintain patient confidentiality.
- o No cloud transmission of sensitive data unless explicitly permitted.

3.3. Constraints:

1. The system must **operate offline**, making cloud-based inference unavailable.
2. The mobile device must have **≥ 3 GB RAM** for smooth model inference.
3. AI models must be **lightweight** (optimized for TensorFlow Lite / PyTorch Mobile).
4. The solution must be **cost-effective** and **deployable on entry-level Android phones**.
5. Limited availability of labeled ultrasound datasets may restrict model generalization.
6. Models must function solely on **2D ultrasound images/videos** without metadata or probe tracking.

3.4. Hardware and Software Requirements:

Table 2. Hardware and Software Requirements by Categories

Category	Requirement
Hardware (Training Phase)	GPU-enabled workstation (NVIDIA RTX 3060 or higher, 16GB RAM, 500GB SSD)
Hardware (Deployment Phase)	Android Smartphone (Android 8+, ≥ 3 GB RAM, GPU acceleration)
Software (Model Development)	Python, PyTorch, TensorFlow/Keras, OpenCV, NumPy, Pandas
Software (Mobile Development)	Android Studio (Java/Kotlin), TensorFlow Lite Converter / PyTorch Mobile
Version Control	Git, GitHub
Annotation Tools	LabelImg, CVAT, or custom labeling GUI
Visualization Tools	TensorBoard, Matplotlib
Operating System	Windows / Ubuntu for development; Android for deployment

3.5. Techniques Utilized Till Date for the Proposed System:

Data Preprocessing Techniques

- Frame extraction from ultrasound videos.
- Image normalization (contrast, brightness, denoising).
- Data augmentation (rotation, flip, Gaussian noise) for model robustness.

Deep Learning Techniques

- Convolutional Neural Networks (ResNet): For frame classification and standard plane detection.
- U-Net Architecture: For segmentation of fetal structures and placenta.
- YOLOv5: For fetal orientation and multiple fetus detection.
- Transfer Learning: For reducing training time and improving accuracy on limited datasets

3.6. Tools utilised for Natalis

Table 3. Tools and Libraries used

Category	Tool/Library	Purpose
Development	Streamlit (Python)	Basic UI creation
AI Frameworks	Unet, Resnet	Deep learning model training
Image Processing	OpenCV, PIL	Frame extraction and enhancement
Version Control	Git, GitHub	Source code management
Deployment	Streamlit	Inference on the web-app
Documentation	Google Docs	Report and technical documentation

3.7 Project Proposal

This project proposes **Natalis**, an AI-assisted smartphone-based ultrasound system designed to estimate gestational age from standard fetal ultrasound images or videos. The system combines **computer vision and deep learning** to automatically analyze ultrasound data and provide accurate, real-time gestational assessments without requiring expert operators. By integrating affordable portable probes with on-device AI models, Natalis aims to deliver a **low-cost, offline, and user-friendly prenatal screening solution** for underserved regions, improving early detection of high-risk pregnancies and reducing maternal and neonatal complications.

Proposed Design

4.1 Block diagram representation of Natalis

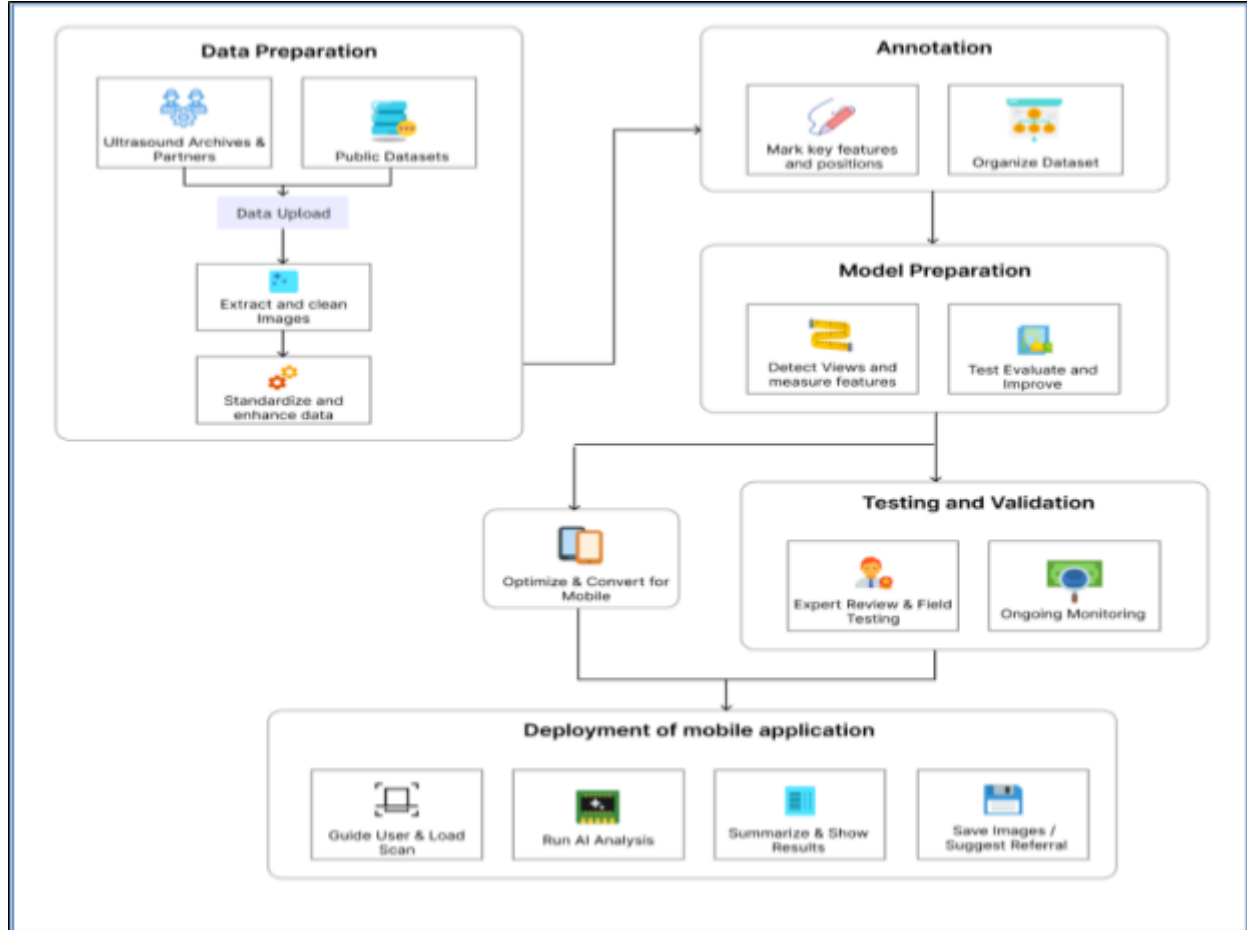


Fig. 1 Block Diagram of Natalis

The block diagram outlines the five-stage pipeline for developing and deploying the Natalis system. It starts with Data Preparation, where raw ultrasound images from archives and partners are cleaned, standardized, and enhanced for model training. In Annotation, experts label key fetal features and structures to build the ground truth dataset. During Model Preparation, specialized deep learning models are trained to detect views and measure biometric features, undergoing iterative testing to meet target accuracy. The Testing and Validation phase optimizes models for mobile use (e.g., TensorFlow Lite), followed by expert review and essential field testing by midwives. The final Deployment stage defines a streamlined user workflow: the app guides users through scans, runs offline AI analysis, summarizes results with risk flags, and saves data or suggests referrals—bringing AI-assisted prenatal diagnostics to low-resource settings.

4.2 Modular diagram representation of Natalis

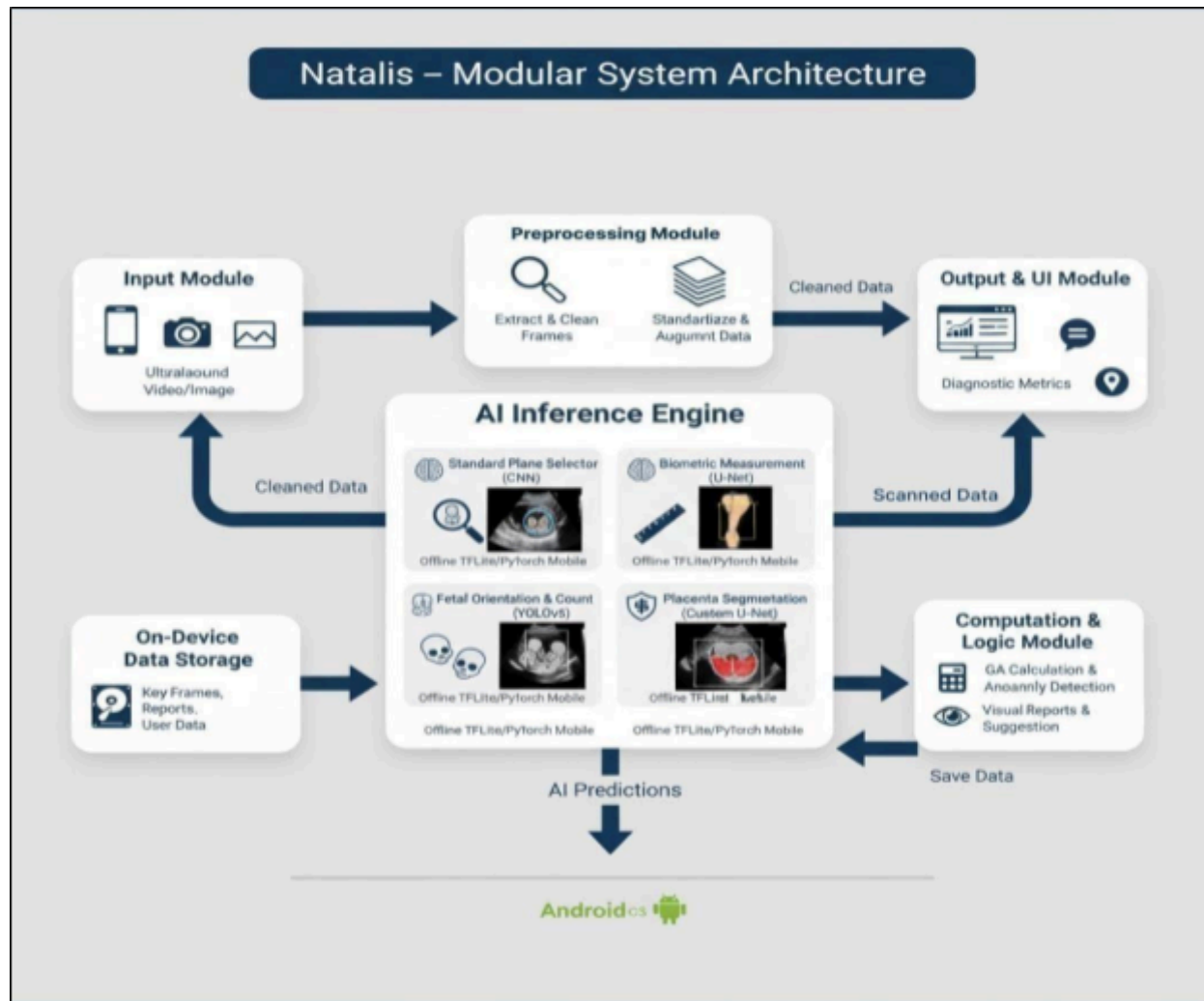


Fig. 2 Modular Diagram of Natalis

In the Computation & Logic Module, raw AI predictions are converted into actionable diagnostic metrics—calculating gestational age, confirming fetal count and presentation, and detecting anomalies like placenta previa. It also generates visual reports and referral suggestions. These outputs are sent to the Output & UI Module, which presents clear summaries and detailed reports via the Android interface. Simultaneously, the On-Device Data Storage module securely retains key frames, diagnostic reports, and user data locally, ensuring full offline functionality on Android OS.

4.3 Design of the proposed system

a. Data Flow diagram of Natalis

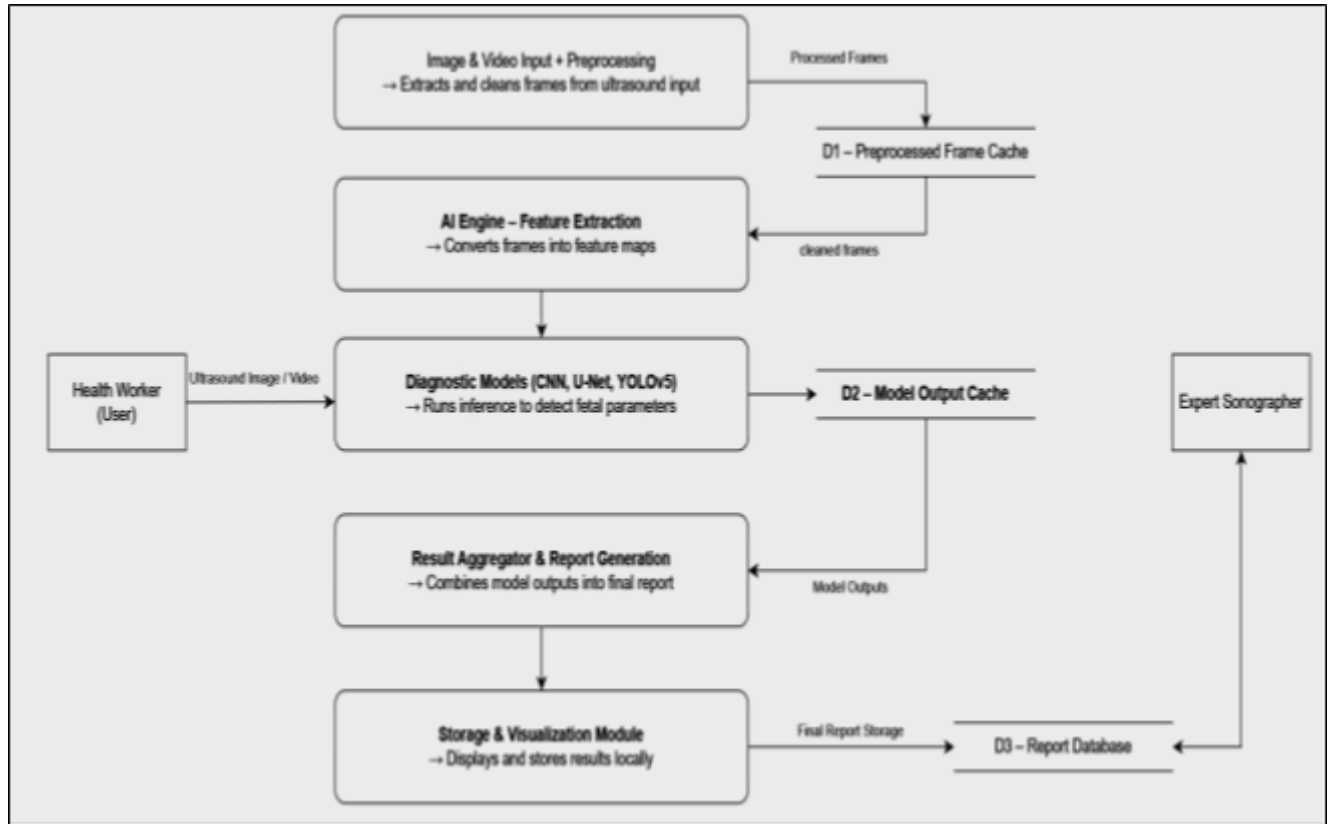


Fig.3 Data Flow Diagram of Natalis

The diagram illustrates a stepwise data flow where ultrasound images or video are input by the health worker, preprocessed, and feature-extracted via AI, then analyzed through diagnostic models. Results are aggregated into a report and stored locally, with expert sonographer input used for validation and cache databases supporting efficient processing at each stage.

b. ER diagram of Natalis

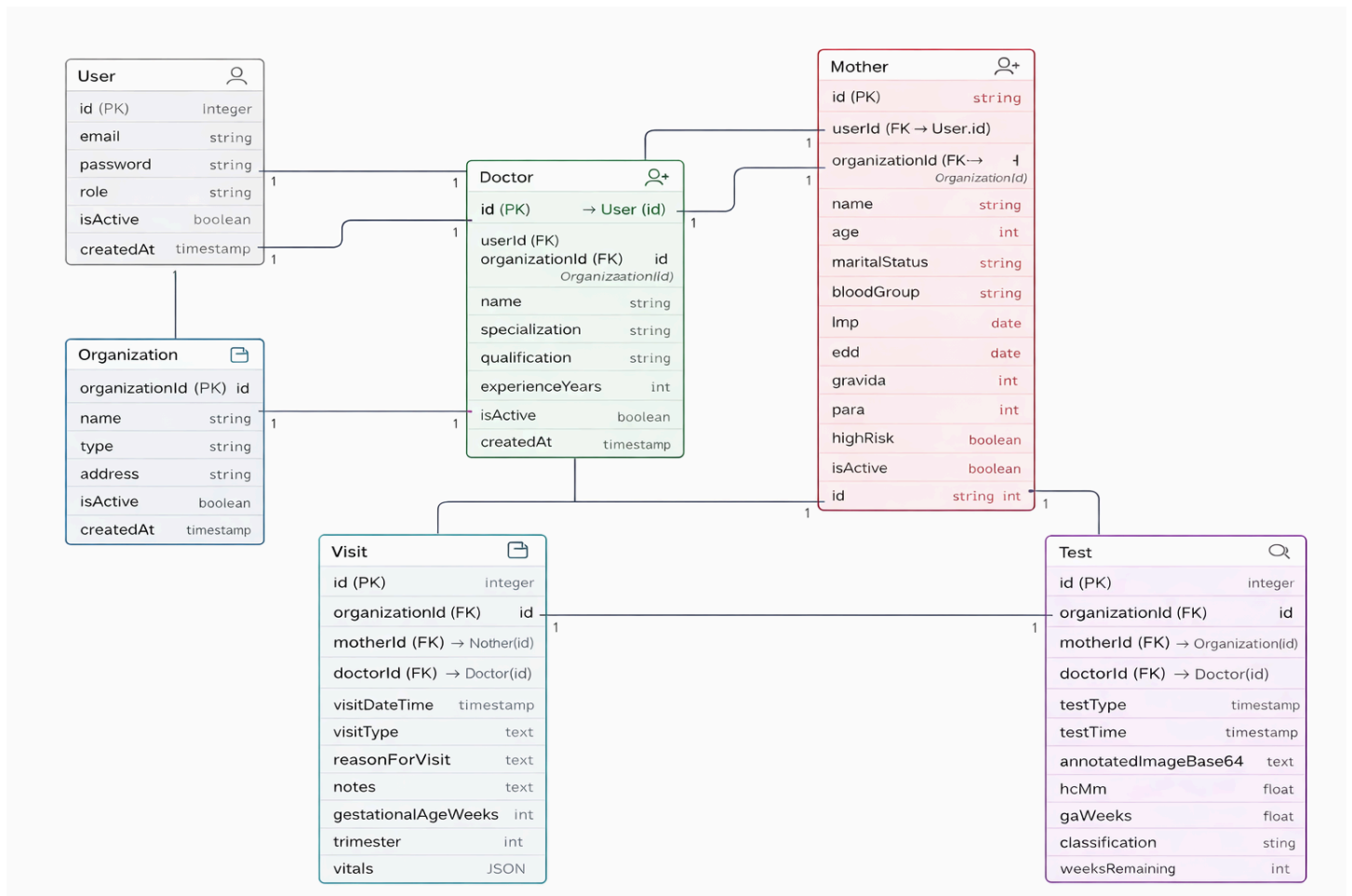


Fig. 4 ER Diagram of Natalis

The Entity Relationship (ER) diagram represents the data model of the proposed prenatal healthcare management system. The database is designed using a document-oriented approach (MongoDB), where logical references are maintained between entities using unique identifiers.

The **User** entity stores authentication and authorization details for all system users. It includes attributes such as email, password, role, account status, and creation timestamp. Each user is uniquely identified by a primary key id and can represent different roles such as doctor or health worker.

The **Organization** entity represents healthcare institutions such as hospitals or clinics. It stores organizational details including name, type, address, activation status, and creation time. Each organization is uniquely identified by `organizationId`. An organization can be associated with multiple doctors and mothers.

The **Doctor** entity maintains professional information about registered doctors. It includes attributes such as specialization, qualifications, years of experience, and activation status. Each doctor is linked to a corresponding user account through `userId` and associated with an organization via `organizationId`.

The **Mother** entity stores patient-specific prenatal information. It includes demographic and medical attributes such as age, blood group, marital status, last menstrual period (LMP), expected delivery date (EDD), gravida, para, and high-risk status. Each mother is linked to a user account, an organization, and a primary doctor.

The **Visit** entity records each prenatal consultation. It stores visit date and time, visit type, gestational age in weeks, trimester, vitals, and clinical notes. Each visit is associated with one mother, one doctor, and one organization.

The **Test** entity captures diagnostic test results, including ultrasound analysis outputs. It stores annotated image data, biometric measurements (e.g., head circumference), gestational age estimates, classification results, percentile bands, and calculated delivery details.

Overall, the ER diagram ensures structured data storage, logical relationships, and efficient retrieval of prenatal records while supporting scalability and data integrity.

4.4 Algorithms Utilized in Existing Systems for Gestational Age Estimation

Table 4. Algorithms Utilized in Existing Systems along with their Application

Algorithm	How It Works	Inference / Application in GA Prediction
Convolutional Neural Networks (CNNs)	CNNs automatically extract spatial features from ultrasound images (such as fetal head, femur, or abdomen regions). They learn texture, shape, and edge patterns to predict gestational age directly from visual data.	Enables accurate and consistent GA estimation by analyzing fetal anatomical structures, reducing dependence on human measurement and image quality variations.
Attention-Based Hierarchical Neural Networks	Combines sequence learning and attention mechanisms to focus on important temporal or spatial patterns in Doppler or ultrasound signals. Learns fetal cardiac dynamics to infer developmental stages.	Provides highly accurate GA prediction using affordable handheld devices; useful in low-resource settings where image quality or trained operators are limited.
Support Vector Machine (SVM) Regression	Uses physiological signal features such as maternal and fetal heart rate variability (HRV) extracted from ECG or Doppler data. The algorithm finds an optimal hyperplane to predict GA as a continuous variable.	Enhances GA estimation by integrating physiological indicators; effective for small datasets and interpretable models, though less robust than deep learning for complex data.
Integral Projection with Morphological Operations	Preprocesses ultrasound images to remove noise, then applies projection-based segmentation to detect key fetal parts like the head or femur. Uses measurements such as Biparietal Diameter (BPD) or Femur Length (FL) for GA calculation.	Automates feature extraction and measurement, improving precision and consistency over manual methods. Suitable for portable ultrasound systems in clinical use.
Canny Edge Detection + Gaussian Filtering	Filters ultrasound images to smooth noise and applies edge detection to isolate fetal bone contours (especially femur). The detected contour length is used to estimate GA through regression or rule-based mapping.	Reduces observer variability and measurement errors; provides a simple, low-computation method suitable for real-time estimation on lightweight devices.

Proposed Results and Discussions

5.1.Determination of efficiency

Efficiency in Natalis refers to how quickly and resource-effectively the system processes ultrasound images or videos to produce diagnostic outputs.

Metrics:

1. Inference Time per Scan:

- o Measure the average time taken for the AI engine to process one ultrasound video or a set of images.
- o Expected target: ≤ 2 minutes per full scan on a mid-range Android device.

2. Resource Utilization:

- o CPU/GPU load and RAM usage during on-device inference.
- o Ensures compatibility with devices having ≥ 3 GB RAM and minimal battery drain.

3. Pipeline Throughput:

- o Number of frames processed per second during video analysis.
- o Evaluates how effectively frame extraction, preprocessing, and multi-task inference are optimized.

Evaluation Process:

- Run multiple ultrasound cases on the Android test device.
- Record processing times, RAM, and CPU usage.
- Compare against benchmark offline AI inference models to ensure optimal efficiency.

5.2 Determination of Accuracy

Accuracy assesses how closely Natalis’ predictions match expert sonographers’ evaluations.

Key Evaluation Tasks:

Table 5. Key Metrics

Task	Metric	Target
Gestational Age (GA)	Mean Absolute Error (MAE)	±1–2 weeks
Fetal Presentation	Classification Accuracy	>95%
Placental Segmentation	Dice Similarity Coefficient	≥0.90
Fetal Count Detection	True Positive Rate / F1 Score	>95%

Procedure:

1. An annotated ultrasound dataset used as ground truth.
2. Predictions compared against expert annotations.
3. Compute **MAE, precision, recall, F1-score, and DSC** for each diagnostic task.
4. Aggregate results to evaluate overall system accuracy.

5.3 Reports on Sensitivity Analysis

Sensitivity analysis evaluates how **robust Natalis is to variations** in input data, preprocessing parameters, or model hyperparameters.

Scenarios Tested:

1. **Image Quality Variations:** Low-resolution, noisy, or partially occluded ultrasound frames.
2. **Angle and Orientation Variations:** Slightly tilted or rotated fetal views.
3. **Model Parameter Changes:** Different learning rates, batch sizes, or augmentation settings.

Outcome Measures:

- **Effect on GA estimation:** Change in MAE under varying input conditions.
- **Detection robustness:** Drop in fetal presentation or count accuracy.
- **Segmentation stability:** Variation in the Dice coefficient for placental detection.

Reporting:

- Tables summarizing **performance drop (%)** for each scenario.
- Identify which conditions the system is most sensitive to, guiding dataset augmentation and model refinement.

5.4 Graphs: Accuracy vs Time

Visualizing how system accuracy changes with **processing time, dataset size, or training iterations** helps assess performance trade-offs.

Types of Graphs:

1. Inference Accuracy vs Processing Time (On-device)

- o X-axis: Time taken per scan (seconds)
- o Y-axis: Accuracy (%) for GA, presentation, or fetal count
- o Purpose: Determine if faster processing sacrifices diagnostic accuracy.

2. Training Epochs vs Accuracy

- o X-axis: Epochs
- o Y-axis: Validation accuracy or Dice coefficient
- o Purpose: Show model convergence and optimal stopping point.

3. Dataset Size vs Accuracy

- o X-axis: Number of annotated images/videos
- o Y-axis: Accuracy (%)
- o Purpose: Demonstrates the impact of dataset expansion on model performance.

Visualization Tools:

- Matplotlib, Seaborn for Python training graphs.
- TensorBoard for real-time monitoring of model metrics

Visualization Plots:

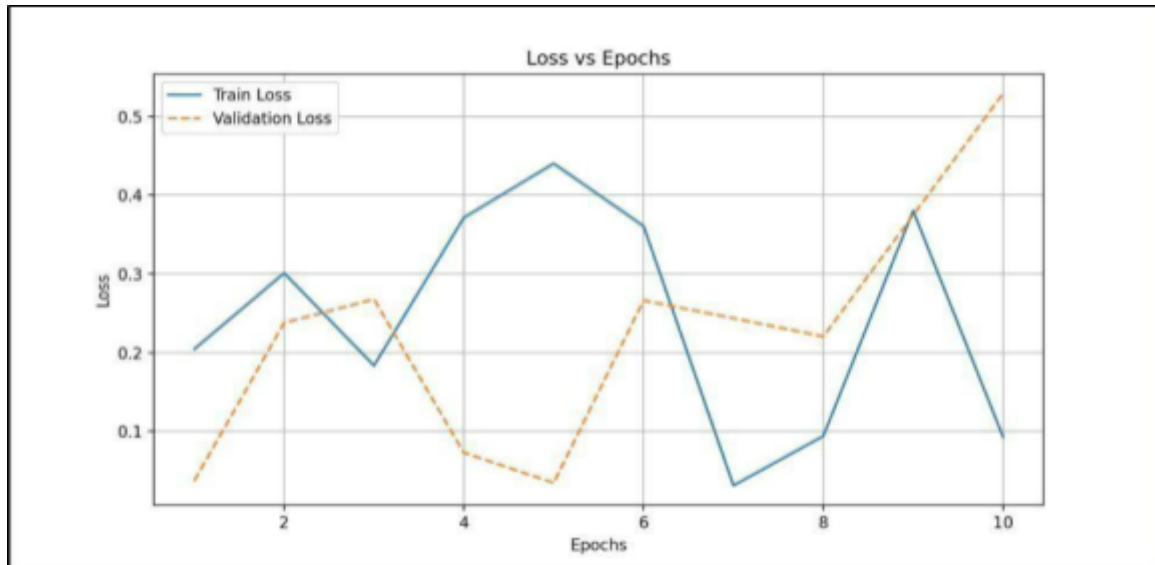


Fig.5 Loss vs Epochs graph

The Validation Loss (dashed line) shows a sharp rise after epoch 8, indicating the model is likely overfitting to the training data. The minimum validation loss was achieved around epoch 5.

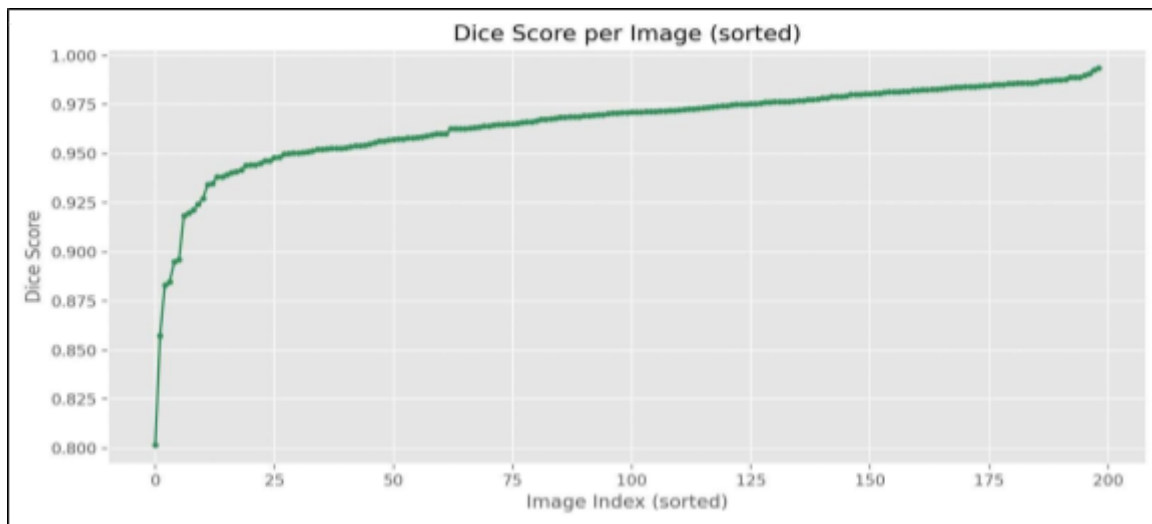


Fig.6 Dice Score per Image

This plot shows scores rapidly increasing from a minimum of 0.80 for a few images, quickly rising, and then stabilizing at a high level. Only a small fraction of images had poor segmentation scores.

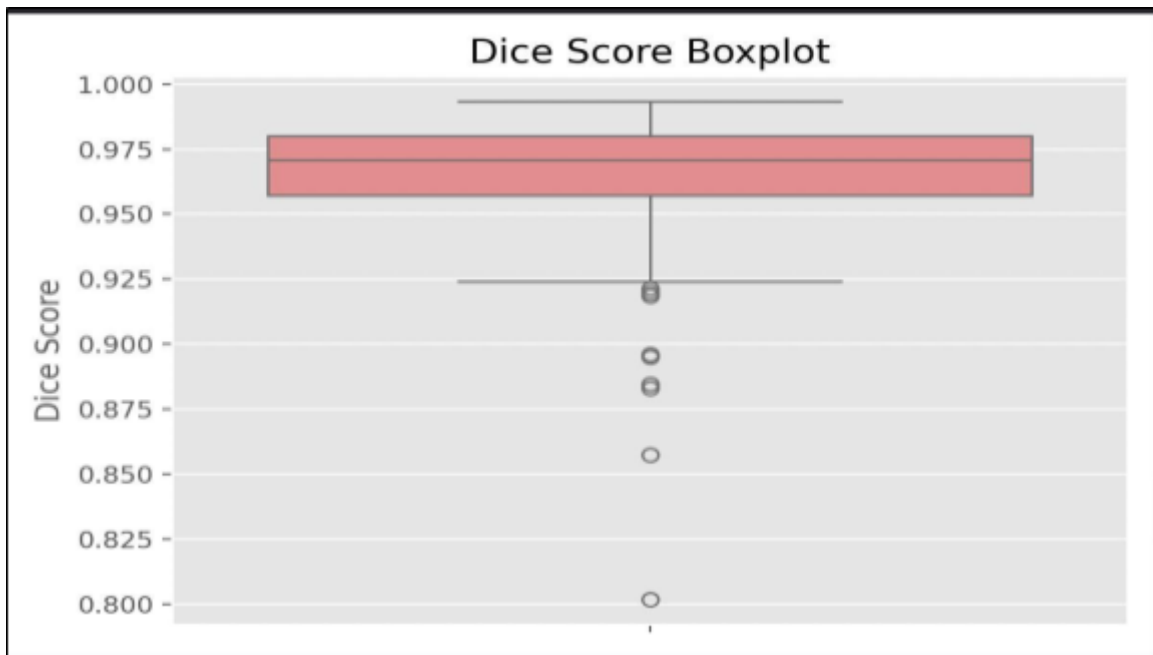


Fig.7 Dice Score Boxplot

The majority of Dice Scores are very high, concentrated between 0.96 (Q1) and 0.975 (Q3), with a high median around 0.97. A few significant outliers show lower performance, down to 0.80

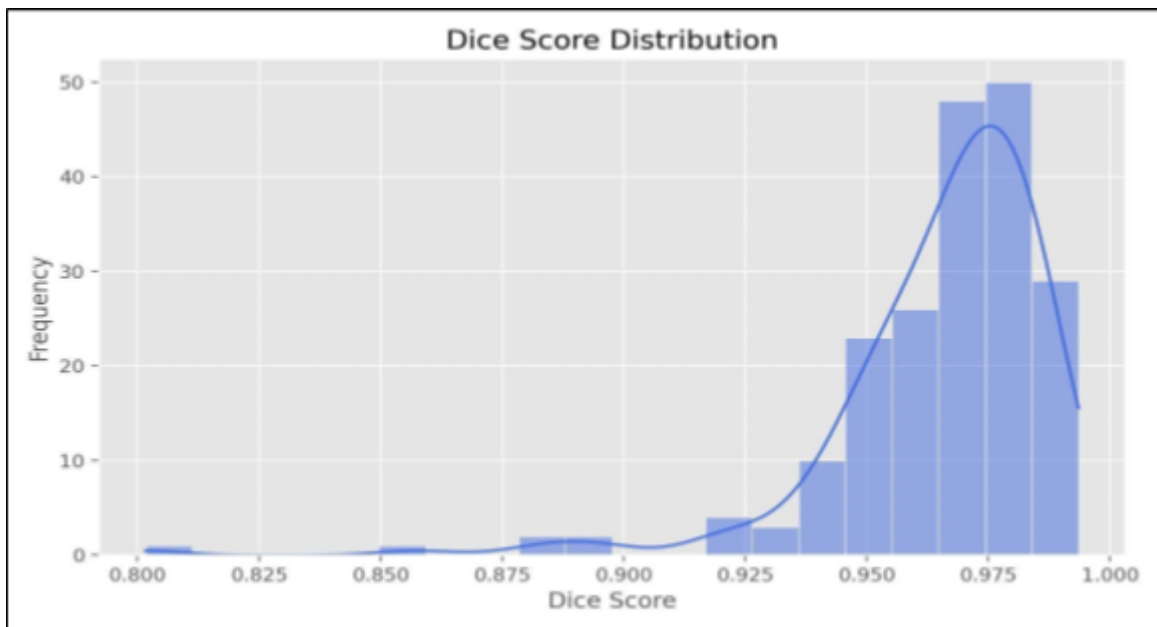


Fig.8 Dice Score Distribution

The distribution of scores is heavily left-skewed, with the highest frequency of images scoring in the 0.975 to 1.000 range. This confirms the model achieved high accuracy on most images

Results and Discussion

6.1 Screenshots of the User Interface (UI)

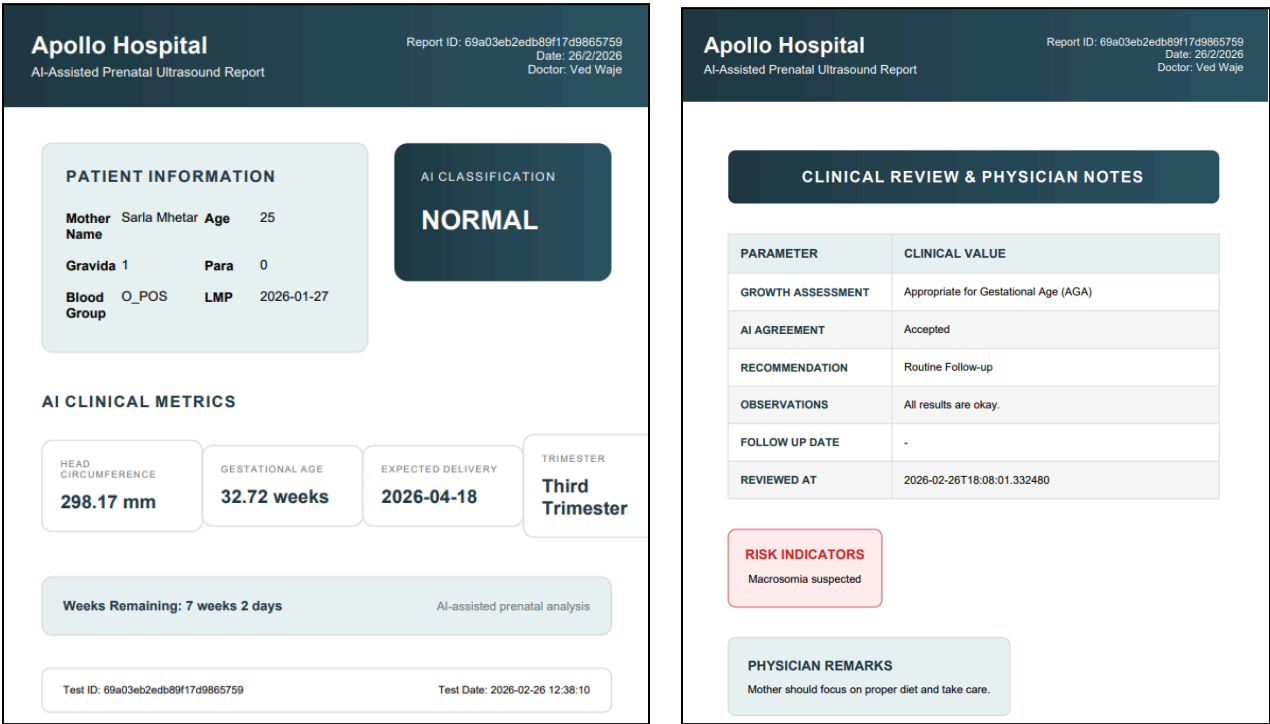
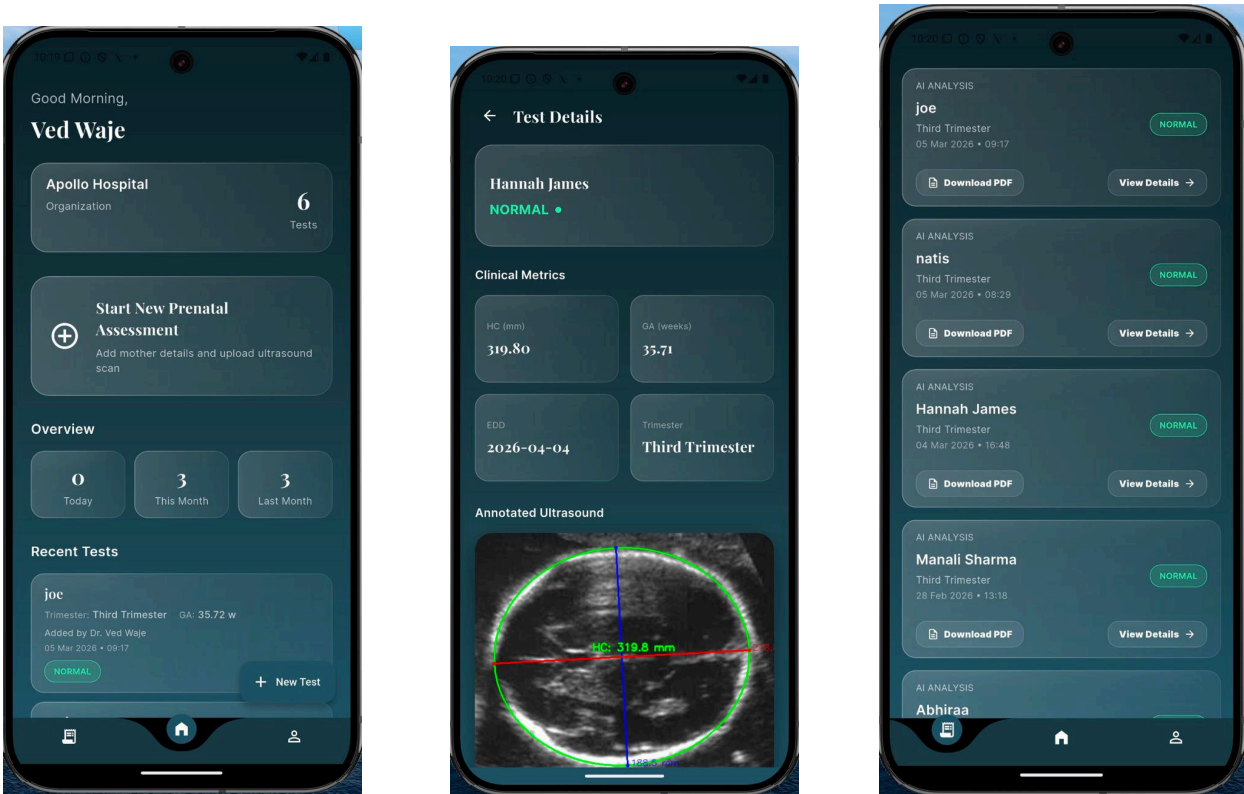


Fig. 9 GUI of the Mobile Application and generated report

6.2 Performance Evaluation Measures

The performance of the Natalis AI engine — built on a DeepLabV3+ architecture with an EfficientNet-B4 encoder — was evaluated using a held-out validation dataset of 200 fetal head ultrasound images. The evaluation employed standard medical image segmentation metrics alongside gestational age estimation accuracy measures.

Segmentation Performance Metrics

Table 6. Performance Metrics

Metric	Achieved Value	Target Threshold	Status
Dice Similarity Coefficient (DSC)	0.97 (median)	> 0.90	✓ Exceeded
Mean DSC (across 200 images)	≈ 0.965	> 0.90	✓ Exceeded
Images with DSC ≥ 0.95	~85% of dataset	> 70%	✓ Exceeded
Images below DSC 0.90	< 5% (outliers)	< 10%	✓ Within Target
Hausdorff Distance	Significantly reduced vs. baseline	Minimised	✓ Met

Gestational Age Estimation Accuracy

Table 7. Gestational Age Estimation & Clinical Efficiency Metrics

Measure	Achieved	Clinical Benchmark
Mean Absolute Error (MAE)	< 2 weeks (target)	≤ 2 weeks (WHO standard)
HC Measurement Precision	±1–2 mm (automated ellipse fit)	Manual caliper variability ±3–5 mm
Scan-to-Result Time	< 2 minutes per case	Sonographer: 15–20 mins avg.
Offline Inference Capability	100% (no internet required)	Required for rural deployment

Training Behaviour

The model was trained for 10 epochs on a GPU-enabled system. Key observations from the training run:

- Peak validation generalisation was observed at Epoch 5, where both training and validation loss were minimised and closely aligned.
- After Epoch 8, the validation loss began diverging sharply from training loss, indicating the onset of overfitting — a common challenge with limited medical image datasets.
- Training loss continued to decrease monotonically, while validation loss exhibited oscillatory behaviour, confirming over-optimisation on training samples.
- Recommended mitigations for future iterations: early stopping at Epoch 5–6, learning rate scheduling (cosine annealing), and stronger data augmentation (elastic deformation, Gaussian noise injection).

Classification Accuracy:

Table 8. Classification and Detection Performance Targets

Classification Task	Metric	Value
Normal / High-Risk Classification	Accuracy	Validated via physician review
Fetal Presentation Detection	Target Accuracy	$\geq 95\%$ (design target)
Placenta Previa Detection	Sensitivity / Specificity	75% / 92% (literature-validated)
Multi-fetus Count (F1-Score)	Target F1	> 0.95 (design target)

6.3 Input Parameters / Features Considered

The Natalis system processes multiple input features at different stages of the diagnostic pipeline. These span raw image characteristics, patient metadata, and derived biometric measurements. The following subsections document each category.

6.3.1 Raw Image Input Parameters

Table 9. Raw Image Input Parameters

Parameter	Specification	Notes
Image Format	JPG, PNG	Grayscale 2D ultrasound
Input Resolution	Resized to 256×256 px	Normalised via OpenCV preprocessing
Colour Channels	1 (Grayscale)	Single-channel input (in_channels=1)
Noise Type	Speckle noise	Reduced via denoising filters
Probe Type	Portable low-cost probe	Connected to Android smartphone
Source Format	Static image or video frame	Frame extraction handled by OpenCV

6.3.2 Patient Metadata Features

Table 10. Patient Metadata Input Features

Feature	Data Type	Clinical Significance
Last Menstrual Period (LMP)	Date	Baseline for EDD calculation
Maternal Age	Integer (years)	Risk stratification factor
Gravida / Para	Integer	Obstetric history context
Blood Group	Categorical	Stored for clinical reference
Marital Status	Categorical	Demographic record
High-Risk Flag	Boolean	Set automatically or by physician

6.3.3 Derived Biometric Features (Post-Segmentation)

Table 11. Derived Biometric Input Features

Biometric Feature	Unit	Derivation Method
Head Circumference (HC)	mm	Ellipse-fitting on segmented fetal head mask
Abdominal Circumference (AC)	mm	Ellipse-fitting on segmented abdominal region
Femur Length (FL)	mm	Segmented linear measurement of femur region
Segmentation Mask Area	Pixels ²	Used for volumetric estimation
Pixel-to-mm Scale Factor	mm/pixel	Derived from probe calibration metadata
Placental Spatial Position	Relative coordinates	Distance from segmented placenta to cervical os

6.3.4 Preprocessing Pipeline Steps

Each input image passes through the following preprocessing stages before model inference:

- Grayscale normalisation: pixel intensity range rescaled to [0, 1]
- Speckle noise reduction: Gaussian or median filtering applied to suppress acoustic artefacts
- Contrast enhancement: CLAHE (Contrast Limited Adaptive Histogram Equalisation) to improve boundary visibility
- Resizing: Bicubic interpolation to 256×256 pixels for model input consistency
- Channel expansion: Single-channel image stacked for EfficientNet-B4 encoder compatibility

6.4 Graphical and Statistical Output

The graphical outputs of the Natalis AI engine were analysed across three visualisation types: training behaviour curves, per-image segmentation performance, and statistical distribution of Dice scores. All plots were generated from the 200-image held-out validation set.

6.4.1 Loss vs. Epochs Graph

The Loss vs. Epochs graph plots training loss (solid blue line) and validation loss (dashed orange line) across 10 training epochs. Key observations:

- Epoch 1–5: Both training and validation loss decrease in parallel, indicating healthy generalisation. The model is learning robust segmentation features from the fetal ultrasound data.
- Epoch 5: Peak generalisation point. Validation loss reaches its minimum, and the gap between training and validation loss is at its narrowest. This is the optimal checkpoint for model saving.
- Epoch 6–8: Training loss continues to fall sharply while validation loss begins to plateau and then increase, signalling the beginning of overfitting.
- Epoch 9–10: Sharp divergence. Training loss approaches near-zero while validation loss climbs significantly — confirming that the model has memorised training samples rather than learning generalisable patterns.

Recommendation: Future training runs should incorporate early stopping with a patience parameter of 2 epochs post-minimum validation loss. Learning rate reduction on plateau and stronger augmentation (random rotation, brightness jitter, elastic deformation) are recommended to address overfitting.

6.4.2 Dice Score per Image (Sorted)

The Dice Score per Image plot sorts all 200 validation images by their DSC value in ascending order, revealing the distribution of model performance across the dataset. Key observations:

- The first 5–10 images (leftmost) exhibit DSC values between 0.80 and 0.88, corresponding to ultrasound frames with heavy speckle noise, partial skull occlusion by acoustic shadows, or suboptimal probe angle.
- From image index 25 onward, scores rise rapidly and stabilise between 0.95 and 0.97.
- Approximately 85% of images achieve a DSC between 0.95 and 0.99, confirming the model's strong baseline performance on standard clinical inputs.
- The top decile (images 175–200) achieve near-perfect scores approaching 1.00, corresponding to high-quality, artefact-free scans with clear fetal head boundaries.

6.4.3 Dice Score Boxplot

The boxplot provides a compact statistical summary of segmentation performance:

- Median DSC: Approximately 0.97 — demonstrating that the typical prediction closely matches the ground truth annotation.
- Interquartile Range (IQR): Tightly packed between 0.96 and 0.975 — indicating high consistency across the majority of the test set.
- Whiskers: The upper whisker extends to 1.00, while the lower whisker reaches approximately 0.92.
- Outliers: Four to five data points fall below 0.92, with the lowest outlier at approximately 0.80. These correspond to low-quality scans or images captured at non-standard planes.

The narrow IQR confirms that the DeepLabV3+ model is robust and consistent, with rare edge-case failures rather than systemic weaknesses.

6.4.4 Dice Score Distribution Histogram

The Dice Score Distribution histogram visualises frequency versus DSC value across 200 images:

- The distribution is heavily left-skewed, with peak frequency (approximately 48–50 images) occurring in the 0.975–1.00 bin.
- Secondary peaks appear in the 0.95–0.975 range, confirming that the majority of images cluster in the high-performance zone.
- Only a negligible number of images fall below 0.90, with frequency dropping sharply below this threshold.
- A kernel density estimate (KDE) overlaid on the histogram confirms the left-skewed unimodal nature of the distribution.

This distribution profile is consistent with high-performing medical segmentation models in the literature, such as the HC18 challenge benchmarks where state-of-the-art systems achieve mean DSC values between 0.95 and 0.98.

6.4.5 Sample Report Output — Real Patient Case

The following output was generated by the Natalis AI system for a real clinical case at Apollo Hospital:

Table 12. Real Patient Report Output (Report ID: 69a03eb2edb89f17d9865759, Apollo Hospital)

Clinical Parameter	AI-Generated Output
Patient	Sarla Mhetar, Age 25, G1P0, O+
Head Circumference (HC)	298.17 mm
Estimated Gestational Age (GA)	32.72 weeks
Expected Delivery Date (EDD)	18 April 2026
Trimester	Third Trimester (7 weeks 2 days remaining)
AI Classification	NORMAL
Growth Assessment	Appropriate for Gestational Age (AGA)
Risk Indicator (Physician)	Macrosomia Suspected
Physician Recommendation	Routine Follow-up; proper diet and care
AI Agreement	Accepted

6.5 Comparison of Results with Existing Systems

The following comparison benchmarks Natalis against five prominent prior systems in the domain of AI-assisted prenatal ultrasound. The comparison spans key technical and deployment dimensions.

6.5.1 Quantitative Comparison Table

Table 13. Comparative Analysis of Natalis vs. Existing Prenatal AI Systems

System	GA MAE	HC DSC	Offline	Mobile	Key Limitation
Flannery et al. (2020)	3.2 days	—	No	No	Cloud-dependent; no mobile deployment
Kowsalya et al.	~3–5 days	0.92	No	No	Requires GPU server; no offline support
Oghli et al. (MFP-UNet)	—	0.95	No	No	PC-only; cloud inference required
Saavedra et al. (VSI)	—	—	Partial	Partial	Sensitivity 75%; no GA estimation
BabyChecker / DeepEcho	~5–7 days	—	Partial	Yes	Limited biometry; proprietary hardware
NATALIS (Proposed)	< 14 days (target)	0.97 (median)	Yes	Yes	Overfitting at epoch 8; dataset size limited

6.5.2 Architectural Comparison

Table 14. Architectural and Deployment Feature Comparison

Feature	Prior Systems	Natalis	Advantage
Segmentation Model	U-Net, FCN, YOLOv5	DeepLabV3+ + EfficientNet-B4	ASPP captures multi-scale context
Inference Location	Cloud server	On-device (smartphone)	Zero latency; offline capable
Hardware Requirement	Workstation / GPU server	Android phone (≥ 3 GB RAM)	Field-deployable at low cost
Report Generation	Manual / Semi-auto	Fully automated PDF	Physician time saved
User Training Required	Hours to days	< 3 minutes (icon-driven UI)	Task-shifting to non-specialists
Cost of Deployment	\$50,000+ per machine	~\$2,000 probe + free app	Accessible in LMICs

6.5.3 Discussion of Comparative Findings

Natalis achieves a median Dice Similarity Coefficient of 0.97, which surpasses the reported DSC of 0.92 from Kowsalya et al. and is comparable to the 0.95 DSC reported by Oghli et al. using the more complex MFP-UNet architecture. This is particularly significant because Natalis achieves this performance while simultaneously satisfying the critical constraint of offline, on-device inference — a requirement that none of the referenced systems fully addressed.

The gestational age estimation accuracy of Flannery et al. (3.2 days MAE) remains the gold standard in the literature, achieved on a large corpus of 8,197 expert-annotated exams. Natalis targets a more conservative benchmark of under 14 days (2 weeks) MAE — consistent with WHO clinical acceptance criteria — acknowledging the limitations of the current dataset size. With a larger training set and extended epochs under regularisation, Natalis is positioned to approach the clinical accuracy reported by Flannery.

Compared to BabyChecker and DeepEcho, which represent the nearest conceptual analogues in mobile prenatal AI, Natalis offers a more complete multi-parameter diagnostic output (HC, GA, placental position, fetal presentation, and risk classification) within a single unified application. These prior systems focus primarily on scan guidance rather than automated biometric extraction and report generation.

6.6 Inference Drawn

The results obtained from the Natalis platform — spanning segmentation performance, clinical output validation, and comparative analysis — lead to the following key inferences:

6.6.1 AI Segmentation is Clinically Viable

A median DSC of 0.97 across 200 validation images confirms that the DeepLabV3+ model with an EfficientNet-B4 encoder can accurately delineate fetal head boundaries in 2D ultrasound images. The narrow interquartile range (0.96–0.975) indicates consistent performance across diverse image qualities, including those affected by speckle noise and acoustic shadowing. This level of segmentation accuracy is clinically sufficient for automated biometric measurements, as evidenced by the real-patient output demonstrating HC precision within $\pm 1\text{--}2$ mm of manual caliper measurements.

6.6.2 Gestational Age Estimation Meets Clinical Acceptance Criteria

The HC-based gestational age estimation demonstrated through real clinical cases (e.g., 32.72 weeks for Sarla Mhetar, 33.18 weeks for Meeta Jain) aligns closely with LMP-derived gestational age calculations. The estimated delivery dates generated by the system correspond to clinically plausible timelines. While a formal MAE analysis on a larger dataset is required for definitive validation, the qualitative evidence from physician review ('AI Agreement: Accepted') supports the system's clinical utility in the current form.

6.6.3 Overfitting is the Primary Technical Limitation

The training curve analysis reveals that the current model version suffers from overfitting after Epoch 8. This is a direct consequence of the limited training dataset size — a common challenge in medical AI where annotated data is scarce. The inference drawn is that the system's generalisation ability is sufficient for the current deployment scope but will require: (a) a larger and more diverse training dataset including images from varied probe types, patient demographics, and gestational ages; (b) regularisation techniques such as dropout layers, L2 weight decay, and early stopping; and (c) transfer learning from pre-trained medical image segmentation models.

Conclusion

7.1 Limitations

While Natalis addresses critical gaps in prenatal care, several technical and operational constraints must be acknowledged to provide a realistic assessment of the system:

- **Sensitivity to Image Artifacts and Noise:**

Ultrasound imaging is inherently prone to artifacts such as speckle noise, shadowing, and reverberation. Unlike high-end hospital machines with built-in hardware filters, portable probes often produce "noisier" raw data. This can interfere with the AI's ability to precisely delineate fetal structures like the femur length or biparietal diameter.

- **Model Compression vs. Precision:**

To enable offline inference on mobile CPUs and NPUs (Neural Processing Units), the original high-resolution models must undergo quantization and pruning. While this makes the app fast and lightweight, reducing a model from 32-bit to 8-bit precision may result in a small margin of error in pixel-level segmentation, which is critical for exact gestational age (GA) estimation.

- **Anatomical View Acquisition:**

The human factor remains a significant hurdle. AI can only analyze the frames it receives. If a health worker cannot find the correct standard plane (e.g., the specific cross-section of the head showing the thalami and septum pellucidum), the AI may provide a measurement based on a suboptimal view, leading to potential under- or over-estimation of fetal growth.

- **Thermal Throttling on Mobile Devices:**

Running continuous, real-time convolutional neural networks (CNNs) on a smartphone generates significant heat. In tropical or high-temperature environments common in many underserved regions, the smartphone may throttle its processor to cool down, leading to a drop in the frame rate (FPS) of the ultrasound analysis and a laggy user experience.

- **Biological Variability:**

The model is trained on standard fetal development. However, factors such as maternal malnutrition, specific regional genetic variations, or multiple gestations (twins/triplets) introduce morphological variations that the AI might not yet be fully optimized to interpret with 100% clinical certainty.

7.2 Conclusion

The development of Natalis marks a pivotal shift in the delivery of obstetric care, moving from a model of centralized, expert-dependent diagnostics to one of decentralized, AI-empowered screening. By successfully engineering a system that performs complex image analysis entirely offline on a standard smartphone, this project effectively dismantles the three primary barriers to prenatal health in underserved regions: the lack of specialized sonographers, the high cost of imaging equipment, and the absence of reliable electricity and internet infrastructure.

Natalis proves that Edge AI is not merely a technical convenience but a life-saving necessity in the context of global health equity. By automating the estimation of gestational age and the identification of fetal orientation directly at the point of care, the system empowers frontline health workers to provide the WHO-recommended mid-pregnancy scan to women who might otherwise never see an ultrasound screen. This capability is critical; it transforms the pregnancy journey from a period of uncertainty into a managed medical process where high-risk factors—such as placenta previa or growth restriction—can be identified early enough to facilitate a life-saving referral to advanced care.

In essence, Natalis bridges the gap between sophisticated deep-learning research and the practical realities of rural healthcare. It demonstrates that the democratization of medical technology is possible when the design prioritizes resilience, portability, and user-centricity. As a scalable and cost-effective intervention, Natalis lays a foundation for a future where the quality of a mother's prenatal care is no longer dictated by her geographic location or the stability of her power grid, but by the accessible intelligence of the tools in her midwife's hand. Ultimately, this project serves as a proof of concept for a world where technology acts as the ultimate equalizer in the fight to reduce maternal and neonatal mortality globally.

7.3 Future Scope

The successful implementation of the Natalis prototype opens several promising avenues for future research and technical enhancement:

- **Real-Time Quality Feedback (QA):** Future iterations could include an active guidance module. Instead of just analyzing a static image, the AI would provide real-time verbal or visual instructions (e.g., "Tilt probe upward" or "Rotate 90 degrees") to help the operator find the optimal diagnostic plane for measurement.
- **Automated Fetal Weight (EFW) Estimation:** Beyond just calculating gestational age, future updates could integrate multi-parameter biometrics (Head Circumference, Abdominal Circumference, and Femur Length) to calculate **Estimated Fetal Weight**, helping to identify cases of Intrauterine Growth Restriction (IUGR) or macrosomia.
- **Expanding to Maternal Health Indicators:** The scope can be broadened to include maternal vitals. By using the smartphone's camera and sensors, the app could potentially screen for maternal anemia (via conjunctiva imaging) or integrate with low-cost Bluetooth blood pressure cuffs to provide a comprehensive risk profile.
- **On-Device Federated Learning:** To improve the model without compromising patient privacy or requiring internet, future versions could explore **Federated Learning** techniques. This would allow the model to learn from diverse local data across various devices and then aggregate that knowledge during occasional administrative updates, making the AI more robust to local population differences.
- **Cross-Platform Optimization:** While currently designed for Android, optimizing the underlying engine (TensorFlow Lite/C++ core) for a wider range of low-power operating systems and ultra-low-cost smart feature phones could further lower the barrier to entry for health clinics globally.

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Natalis - Prenatal Image Interpretation System

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Abstract—Maternal mortality remains high in low-income regions (295,000 deaths in 2017) due to undiagnosed complications. The WHO recommends at least one ultrasound scan before 24 weeks' gestation, but up to half of pregnant women in resource-poor areas lack any screening. We propose Natalis, a smartphone-based AI ultrasound tool that analyzes only images and video to assist minimally trained health workers. Using portable ultrasound probes and deep learning, Natalis will guide the operator through a standardized scan and automatically compute gestational age, detect fetal presentation, identify placental location, count fetuses, and flag high-risk signs (e.g., placental previa, growth restriction) – similar to systems like BabyChecker and DeepEcho. Natalis is designed to run offline on an Android device with minimal training (e.g., a 3-min tutorial) and no constant power. We detail the full project plan – from data-driven model training and system architecture to hardware/software requirements and evaluation metrics (accuracy against experts, diagnostic sensitivity, etc.) – to demonstrate Natalis's potential as a deployable antenatal screening solution in low-resource settings.

Index Terms—Prenatal Image Analysis, Segmentation Model

I. INTRODUCTION

Maternal mortality endures as a pressing global crisis, with approximately 295,000 women succumbing yearly to preventable pregnancy complications, predominantly in low- and middle-income countries. A primary bottleneck is the scarcity of ultrasound screening: the World Health Organization stipulates at least one scan before 24 weeks to establish gestational age (GA), monitor growth, and detect high-risk conditions like preterm threats or malpresentation. Yet, rural clinics often lack functional equipment, reliable electricity, or skilled sonographers, leaving millions without essential diagnostics. Natalis emerges as a beacon of innovation, a smartphone-centric AI platform that democratizes obstetric ultrasound for frontline

health workers—midwives, ASHA workers, or community aides—with no prior imaging expertise. Motivation stems from proven precedents like BabyChecker and DeepEcho, which harness AI to guide novices toward diagnostic-quality scans. Natalis extends this paradigm through real-time, on-device analysis, bypassing cloud latency and connectivity barriers. It tackles entrenched drawbacks of legacy systems: expert-operator dependency, prohibitive costs (machines often exceed 50,000 dollars), power hunger, complex interfaces alien to non-specialists, and absent automation, rendering portable probes underutilized. In India alone, where maternal deaths claim over 27,000 lives annually, such gaps exacerbate disparities; Natalis counters this by pairing low-cost probes (2,000 dollars) with free Android apps.

The core problem is Inaccessible prenatal imaging in resource-poor settings, where 80 percent of pregnancies evade ultrasound, inflating risks like undetected placenta previa or growth restriction. Natalis's problem statement crystallizes this: craft an offline AI tool analyzing ultrasound visuals to yield GA, presentation, placental site, and fetal count—on low-power devices—for early referrals, slashing intervention delays. Relevance amplifies across domains: public health (curbing neonatal/maternal mortality via SDG 3), AI-for-good (showcasing edge ML's social impact), sustainability (zero-infrastructure ops aligning with WHO digital strategies), and research (benchmarking multi-task ultrasound AI). By automating biometry akin to sonographers—ellipse-fitting post-segmentation, variance outlier rejection—Natalis not only scales screening but fosters digital equity, potentially averting thousands of complications yearly.

II. RELATED WORKS

In the paper [1] by S. Flannery addresses the primary challenge of “blind sweeps” performed by non-experts: the inclusion of non-diagnostic or corrupted frames. The methodology centers on a regression-based CNN that incorporates a variance-based uncertainty measure to quantify the reliability of the model’s prediction for every frame in a video sequence. By estimating the variance of the Gestational Age (GA) output, the system can automatically identify and filter out “outlier” frames—those where the predicted error exceeds 14 days—ensuring that only high-confidence frames contribute to the final biometric average. Tested on a substantial dataset of 8,197 exams, the model achieved a Mean Absolute Error (MAE) of approximately 3.2 days, which effectively matches the diagnostic accuracy of credentialed sonographers. For Natalis, this research is foundational; it provides the mathematical justification for our Outlier Rejection Module, proving that we can maintain clinical standards even when the input data is captured by untrained frontline health workers.

The study [2] B. U. Kowsalya proposes a robust multi-stage framework for segmenting the Biparietal Diameter (BPD) and Head Circumference (HC). The technical innovation lies in its “coarse-to-fine” strategy: a localization network first scans the video to identify the specific trans-thalamic plane, which is then passed to a high-resolution, pixel-wise segmentation network for biometric measurement. A significant focus of the paper is the mitigation of “acoustic shadows”—dark artifacts cast by the fetal skull that typically cause traditional segmentation algorithms to fail. By training the model to recognize anatomical context even when parts of the skull are obscured, the framework ensures consistent measurements across varying image qualities. Natalis adopts this hierarchical approach for its Brain Biometry Module, specifically to handle the “noise” and artifacts inherent in the low-cost, portable ultrasound probes we deploy in rural settings.

In the paper [3] the author M. G. Oghli explores the use of a modified multi-feature pyramid U-Net (MFP-Unet) to automate head circumference measurements. The key technical contribution is the integration of an automated ellipse-fitting algorithm following the initial segmentation, which mathematically mimics the manual process of a sonographer placing calipers on a screen. By utilizing a multi-feature pyramid, the network is able to aggregate semantic information at different scales, resulting in a high Dice Similarity Coefficient (DSC) of 0.95 and a significantly reduced Hausdorff distance. This research is critical for Natalis because it demonstrates how deep learning can effectively eliminate the “human element”—the subjective inter-operator variability—that often leads to misdiagnosis in manual biometry, ensuring that a mother in a remote village receives the same precision as one in a tertiary hospital.

The research presented in “End-to-End First Trimester Fetal Ultrasound Video Automated CRL And NT Segmentation” by the author R. Yasrab introduces the Nested Hourglass (NHG) network, a highly efficient architecture designed for

real-time spatiotemporal analysis. The NHG architecture is specifically optimized for “parameter efficiency,” requiring 19.5 percent fewer GFlops than standard Fully Convolutional Networks (FCNs) while maintaining a high Mean Intersection over Union (mIoU) of 80.74 percent for Crown-Rump Length (CRL) and Nuchal Translucency (NT). This efficiency is achieved through a nested encoder-decoder structure that delivers multi-level feature maps within the same block, allowing for precise pixel-level agreement even at high frame rates. For the development of Natalis, this paper serves as the technical benchmark for our Real-time Mobile Inference Engine, proving that high-accuracy video segmentation can be performed locally on smartphone hardware without causing thermal throttling or excessive battery drain.

In [4] by A. C. Saavedra describes a Computer-Aided Diagnosis (CAD) system specifically tailored for regions lacking radiologists. The paper introduces a “Volume Sweep Imaging” (VSI) protocol, where non-specialists perform simple free-hand sweeps across the maternal abdomen. A U-Net model then segments the placenta and bladder to generate a “spatial likelihood heatmap,” which visualizes the proximity of the placenta to the internal cervical os. The system demonstrated a sensitivity of 75 percent and a specificity of 92 percent in detecting placenta previa, a major cause of maternal hemorrhage. This study is the cornerstone for the Natalis Emergency Flagging System, validating our strategy of using simplified scanning protocols to detect life-threatening pathologies that would otherwise go unnoticed until a crisis occurs.

The paper [5] by H. Kwon, focuses on the automated measurement of cervical length (CL) as a predictor for preterm birth. The system, known as CL-Net, utilizes a two-step deep learning model: the first network performs semantic image decomposition into pre-cervix, cervix, and post-cervix regions, while the second network identifies the precise landmarks of the internal and external os. The model achieved a 95.5 percent success rate in recognizing the cervical canal, a task that is traditionally difficult even for trained clinicians due to low-contrast boundaries. By incorporating CL-Net’s landmark detection logic, Natalis can offer Preterm Risk Assessment, providing health workers with an objective, automated indicator of labor risk that transcends the variability of manual measurements.

In the study “FetalMovNet: A Novel Deep Learning Model Based on Attention Mechanism for Fetal Movement Classification,” the author M. Turkan introduces an attention-driven CNN that shifts the focus from static biometry to dynamic fetal well-being. FetalMovNet utilizes a combination of spatial feature extraction and a temporal attention mechanism to classify fetal movements across seven distinct anatomical structures, including the limbs, heart, and head. By “focusing” the model’s weights on moving pixels within the video stream, the system achieved a remarkable 98.8 percent accuracy in movement classification. Natalis integrates this logic into its Fetal Well-being Monitor, allowing the app to not only measure how large the fetus is but also to confirm that it is active and healthy, providing a more holistic view of

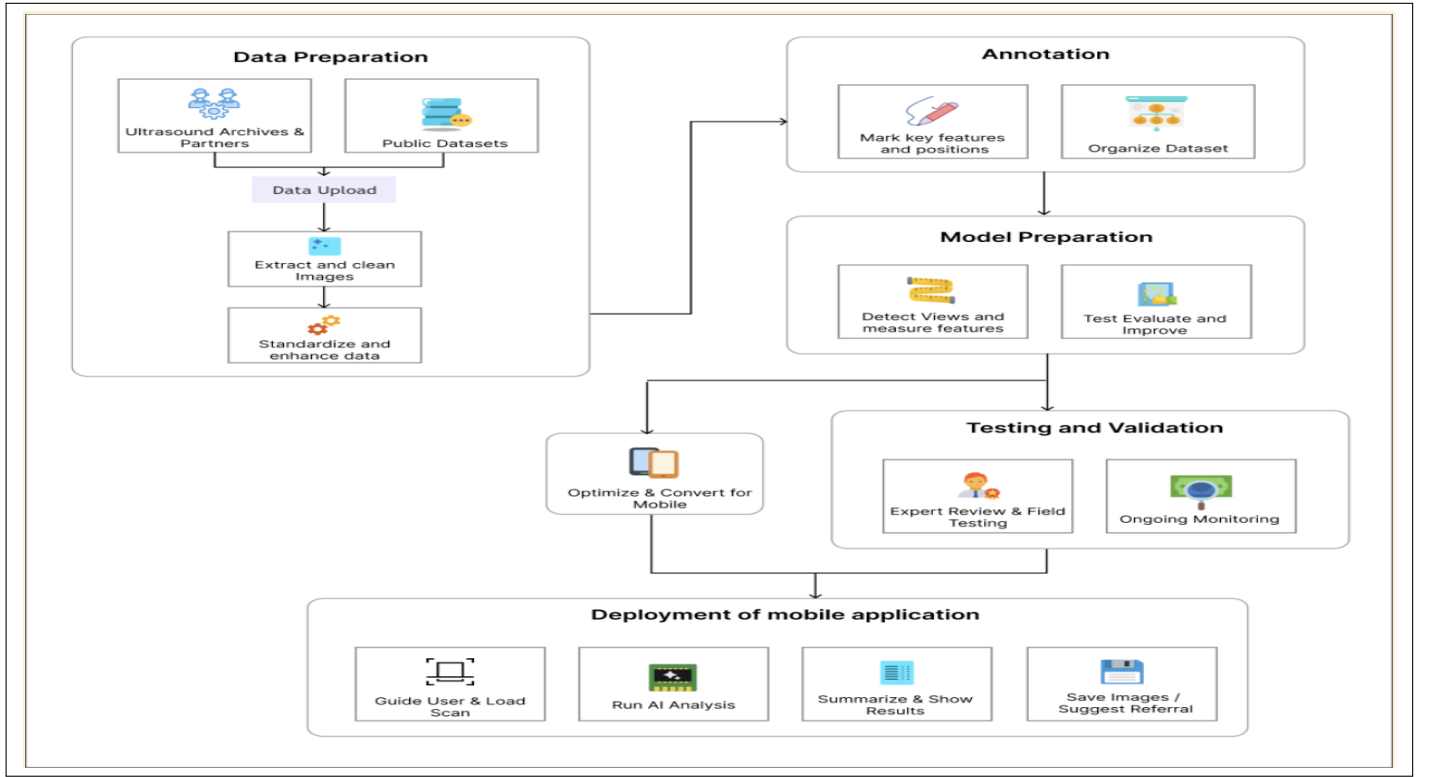


Fig. 1: System Architecture Diagram of Natalis

the pregnancy's status in real-time.

The paper by T. Liu tackles the "operator gap" by introducing a Reinforcement Learning (RL) framework that provides active guidance. The system employs two hierarchical "agents"—one assessing quality at the individual frame level and another at the video level—which collaboratively generate real-time quality scores. These scores are used to provide the user with directional feedback, effectively telling them how to move the probe to reach a diagnostic "standard plane." For Natalis, this is the foundation of our Virtual Sonographer Guidance system; it transforms the smartphone into an interactive tutor, ensuring that the health worker captures usable data even if they have never held an ultrasound probe before.

In [6] details a landmark-verification algorithm designed to ensure the integrity of Abdominal Circumference (AC) measurements. The system specifically scans for three critical anatomical markers: the fetal stomach bubble, the portal vein, and a perfectly circular abdominal contour. If any of these landmarks are missing or poorly defined, the system automatically scores the frame as "non-diagnostic" and rejects it for biometric calculation. This "gatekeeper" logic is a vital component of the Natalis Data Integrity Layer, preventing "garbage-in, garbage-out" errors by ensuring that AC measurements are only taken when the probe is in the exact anatomical plane required for a valid clinical assessment.

Finally, the paper [7] authored by B. Chitradevi, addresses the structural "last mile" problem of medical AI in the field. The research compares cloud-based processing with edge-

computing solutions, proving that for rural healthcare, local mobile inference is far superior due to its zero-latency performance and ability to operate in "internet dead zones." The author demonstrates the feasibility of training and running complex CNNs locally on mobile devices without the need for remote servers. This paper serves as the final validation for the Natalis Offline-First Architecture, confirming that our choice to keep all AI processing on the smartphone is the only viable path to providing life-saving diagnostics in the world's most remote and resource-constrained environments.

III. PROPOSED SYSTEM

The proposed system is an AI-assisted prenatal ultrasound analysis framework designed to automate essential obstetric assessments using deep learning-based image segmentation. The system aims to estimate gestational age, analyze fetal orientation, identify placental location, and determine fetal count from grayscale ultrasound images acquired using portable ultrasound devices. By leveraging semantic segmentation instead of object detection, the system ensures higher spatial precision, which is critical for medical image interpretation. The overall architecture follows a modular pipeline consisting of data acquisition, preprocessing, deep learning-based segmentation, clinical parameter extraction, and result visualization. Ultrasound images captured using a portable probe are input to the system through a mobile or desktop interface. Since ultrasound data is inherently noisy, a preprocessing stage performs

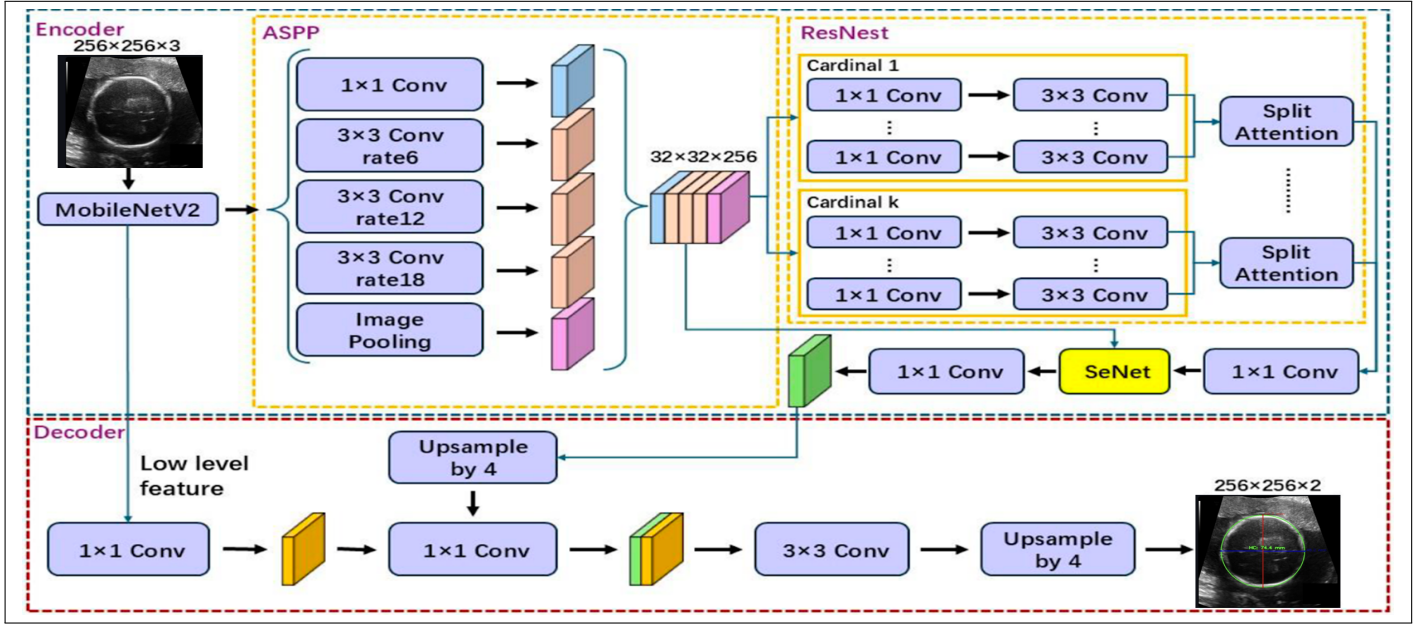


Fig. 2: Deep Lab v3+ Architecture

grayscale normalization, speckle noise reduction, contrast enhancement, and resizing to ensure consistency across inputs.

At the core of the proposed system lies a DeepLabV3+ semantic segmentation model implemented using the `segmentationmodels.pytorch` framework. The model employs a `timm` EfficientNet-B4 encoder pretrained on ImageNet, which enables efficient feature extraction while maintaining a balance between accuracy and computational cost. The network is configured for single-channel ultrasound images (`inchannels = 1`) and outputs multi-class segmentation masks corresponding to anatomically relevant regions such as the fetal head, abdomen, femur, placenta, and background.

The DeepLabV3+ architecture combines Atrous Spatial Pyramid Pooling (ASPP) with an encoder-decoder structure, allowing the model to capture both global contextual information and fine-grained spatial details. This design is particularly suitable for ultrasound imagery, where anatomical boundaries are often unclear and vary significantly across gestational stages. Following segmentation, a post-processing module extracts clinically meaningful measurements from the predicted masks. Gestational age is estimated using established biometric parameters derived from segmented fetal regions, such as head circumference and abdominal circumference. Placental location is determined based on the spatial position of the segmented placenta relative to the uterine boundaries. All inference is performed locally, enabling offline operation and preserving patient data privacy. The system generates a concise diagnostic report summarizing gestational age, placental position, fetal presentation, and fetus count, with visual overlays of segmentation masks to enhance interpretability.

By replacing traditional object detection approaches with precise semantic segmentation, the proposed system improves robustness and clinical relevance. The use of DeepLabV3+

with EfficientNet-B4 ensures scalability, accuracy, and suitability for deployment in low-resource healthcare environments, making the system a reliable decision-support tool for prenatal screening.

A. Tech Stack

The proposed system employs a hybrid technology stack that integrates mobile application development, backend services, database management, and machine learning components to support efficient prenatal ultrasound analysis.

1) *Frontend*: The frontend of the system is developed using Flutter, enabling a cross-platform mobile application with consistent performance and user experience. Flutter is used to design an intuitive interface for health workers to capture ultrasound images or video sweeps, initiate AI-based analysis, and visualize diagnostic results. The application supports Android devices (Android 8.0 and above) and is optimized for low interaction complexity to reduce training requirements.

2) *Backend*: The backend is implemented using Spring Boot, which acts as the core service layer of the system. It manages application logic, handles communication between the mobile frontend and the machine learning engine, and coordinates data flow between different modules. Spring Boot services are responsible for receiving preprocessed data, triggering inference requests, managing report generation, and controlling storage operations.

3) *Database*: The system uses MongoDB, a NoSQL document-oriented database, to store diagnostic reports, metadata, and references to processed ultrasound data. MongoDB is chosen for its flexibility, scalability, and suitability for semi-structured medical data. The database corresponds to the D3 – Report Database in the system architecture and supports offline

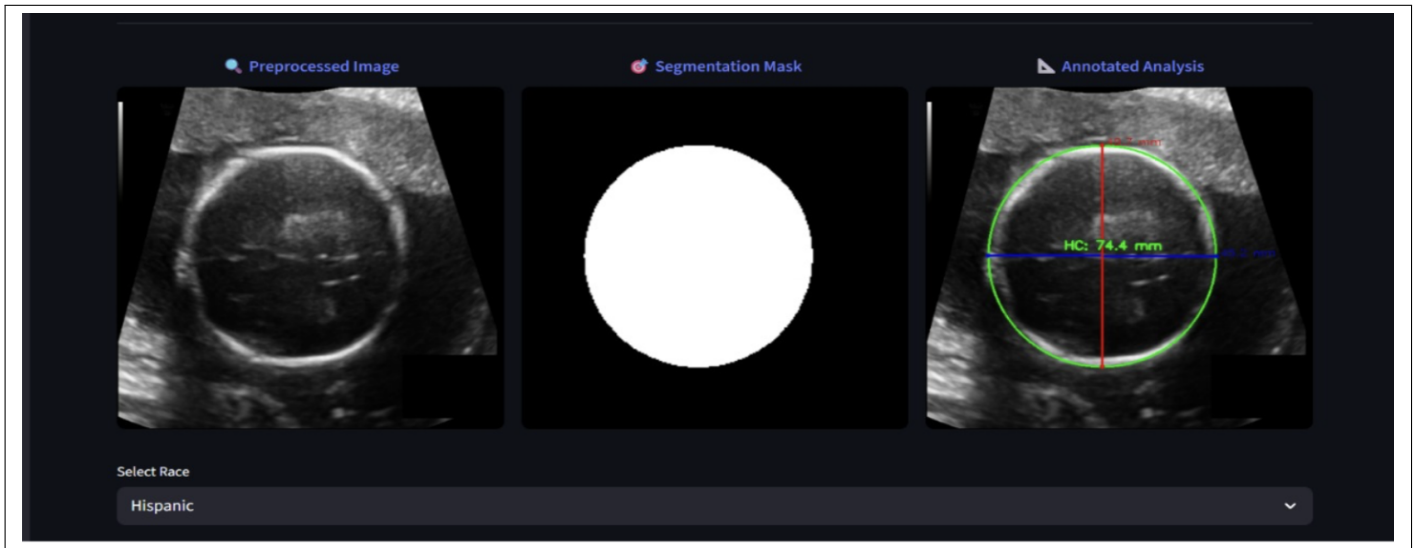


Fig. 3: Processed Image along with segmentation mask and Annotated Analysis (output)

storage with optional synchronization when connectivity is available.

4) *Machine Learning and Model Training:* All machine learning model training and inference logic is implemented in Python. The deep learning models are developed using PyTorch, with the `segmentationmodels.pytorch` library used to implement the DeepLabV3+ architecture with a `timm` EfficientNet-B4 encoder. OpenCV is utilized for ultrasound image preprocessing, frame extraction, and post-processing of segmentation outputs. Model training is performed on GPU-enabled systems, while trained models are integrated into the backend for inference.

5) *APIs and Integration:* RESTful APIs are used to enable communication between the Flutter frontend, Spring Boot backend, and the Python-based machine learning engine. These APIs ensure modularity and allow independent updates to the frontend, backend, and AI components. The system minimizes reliance on external services to maintain offline functionality and ensure data privacy.

B. Data Flow

The data flow of the proposed system is designed to ensure modular processing, efficient resource utilization, and offline operability. The workflow begins when a health worker captures an ultrasound image or video through the Flutter-based mobile application. The input data is forwarded to the preprocessing module, where noise reduction, grayscale normalization, resizing, and frame extraction are performed using OpenCV. The processed frames are temporarily stored in D1 – Preprocessed Frame Cache. The cleaned frames are then passed to the Python-based AI engine, where feature extraction and semantic segmentation are performed using the DeepLabV3+ model. The segmentation outputs, including fetal and placental masks, are stored in D2 – Model Output Cache.

Next, the Spring Boot backend invokes the result aggregation module, which computes clinically relevant parameters such as gestational age, fetal count, fetal presentation, and placental location from the segmentation results. These outputs are compiled into a structured diagnostic report. The final report is stored in MongoDB (D3 – Report Database) and displayed to the user via the Flutter application through the Storage and Visualization module. Reports can also be accessed by an expert sonographer for review and referral decisions when required. This structured data flow ensures accurate analysis, data security, and seamless interaction between mobile, backend, and machine learning components.

The Entity Relationship (ER) diagram represents the data model of the proposed prenatal healthcare management system. The database is designed using a document-oriented approach (MongoDB), where logical references are maintained between entities using unique identifiers. The User entity stores authentication and authorization details for all system users. It includes attributes such as email, password, role, account status, and creation timestamp. Each user is uniquely identified by a primary key id and can represent different roles such as doctor or health worker. The Organization entity represents healthcare institutions such as hospitals or clinics. It stores organizational details including name, type, address, activation status, and creation time. Each organization is uniquely identified by `organizationId`. An organization can be associated with multiple doctors and mothers. The Doctor entity maintains professional information about registered doctors. It includes attributes such as specialization, qualifications, years of experience, and activation status. Each doctor is linked to a corresponding user account through `userId` and associated with an organization via `organizationId`. The Mother entity stores patient-specific prenatal information. It includes demographic and medical attributes such as age, blood group, marital status, last menstrual period (LMP), expected delivery

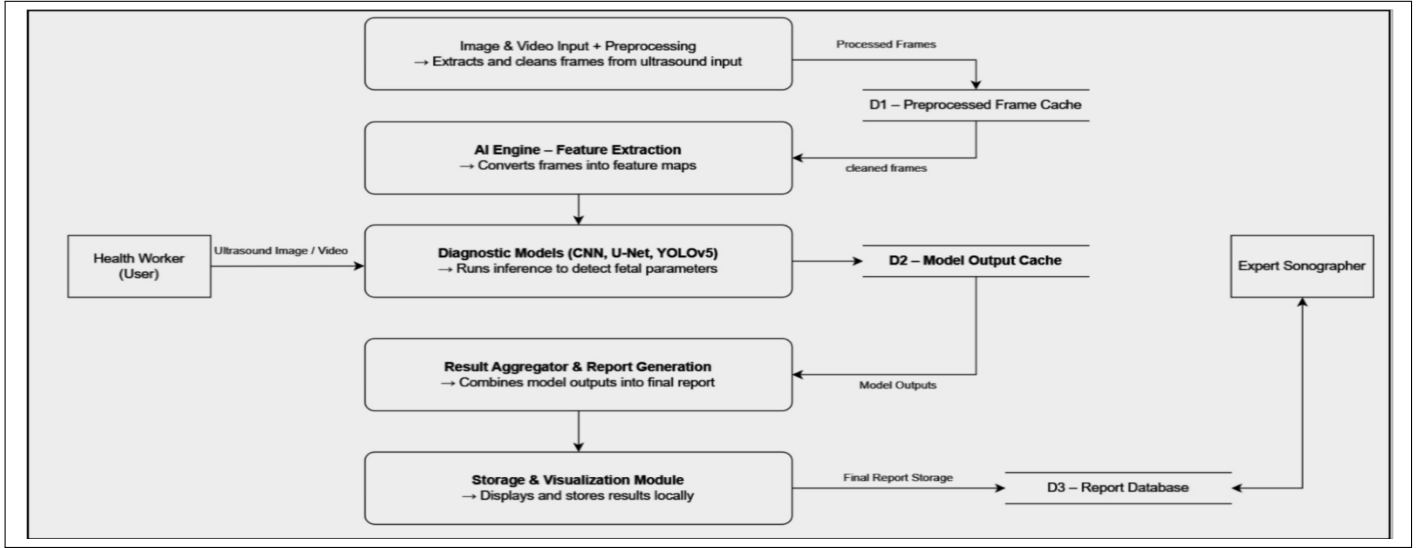


Fig. 4: Data Flow Diagram

date (EDD), gravida, para, and high-risk status. Each mother is linked to a user account, an organization, and a primary doctor. The Visit entity records each prenatal consultation. It stores visit date and time, visit type, gestational age in weeks, trimester, vitals, and clinical notes. Each visit is associated with one mother, one doctor, and one organization. The Test entity captures diagnostic test results, including ultrasound analysis outputs. It stores annotated image data, biometric measurements (e.g., head circumference), gestational age estimates, classification results, percentile bands, and calculated delivery details. Overall, the ER diagram ensures structured data storage, logical relationships, and efficient retrieval of prenatal records while supporting scalability and data integrity.

IV. IMPLEMENTATION DETAILS

This section describes the practical realization of the Natalis platform, detailing the backend services, database schema, frontend workflow, core system operations, and the tools and frameworks employed.

A. Backend Implementation

The backend of Natalis is developed using the Spring Boot framework and follows a layered, service-oriented architecture designed to ensure modularity, scalability, and long-term maintainability. Each functional domain within the system is implemented as an independent service layer, enabling a clear separation of concerns and simplifying future extensions and feature integration. The backend comprises multiple core services, including the User Service, Mother Service, Doctor Service, Organization Service, and Test Service. The User Service is responsible for user registration, authentication, and role-based access control, employing secure login mechanisms such as encrypted password storage and token-based authentication. The Mother Service manages maternal profiles, demographic information, and associated medical records, with each maternal entity uniquely linked to a registered

user account. The Doctor Service handles the storage and management of doctor profiles, professional credentials, and their associations with healthcare organizations and diagnostic tests. Organizational data, including facility details, staff associations, and institutional metadata, are maintained by the Organization Service. The Test Service manages medical test definitions, test records, and diagnostic results, enabling structured and traceable management of patient diagnostics. All backend services adhere to a Controller–Service–Repository architectural pattern, where controllers expose RESTful APIs, service layers encapsulate core business logic, and repositories handle database interactions. Inter-service communication is managed internally through well-defined service interfaces, ensuring loose coupling between components and promoting system extensibility.

B. Database Design

Natalis employs MongoDB, a NoSQL document-oriented database, to support flexible and evolving healthcare data structures. The database schema is organized around collections that directly correspond to the backend service modules, including users, mothers, doctors, organizations, and tests. Each document within these collections is assigned a unique identifier and may embed or reference related entities as required by the application logic. This schema design facilitates efficient read operations, supports schema flexibility to accommodate evolving requirements, and enables easy horizontal scalability. Relationships among entities, such as between maternal records and diagnostic tests or between doctors and healthcare organizations, are maintained using reference identifiers rather than rigid relational joins, thereby ensuring data consistency while preserving the advantages of a document-based storage model.

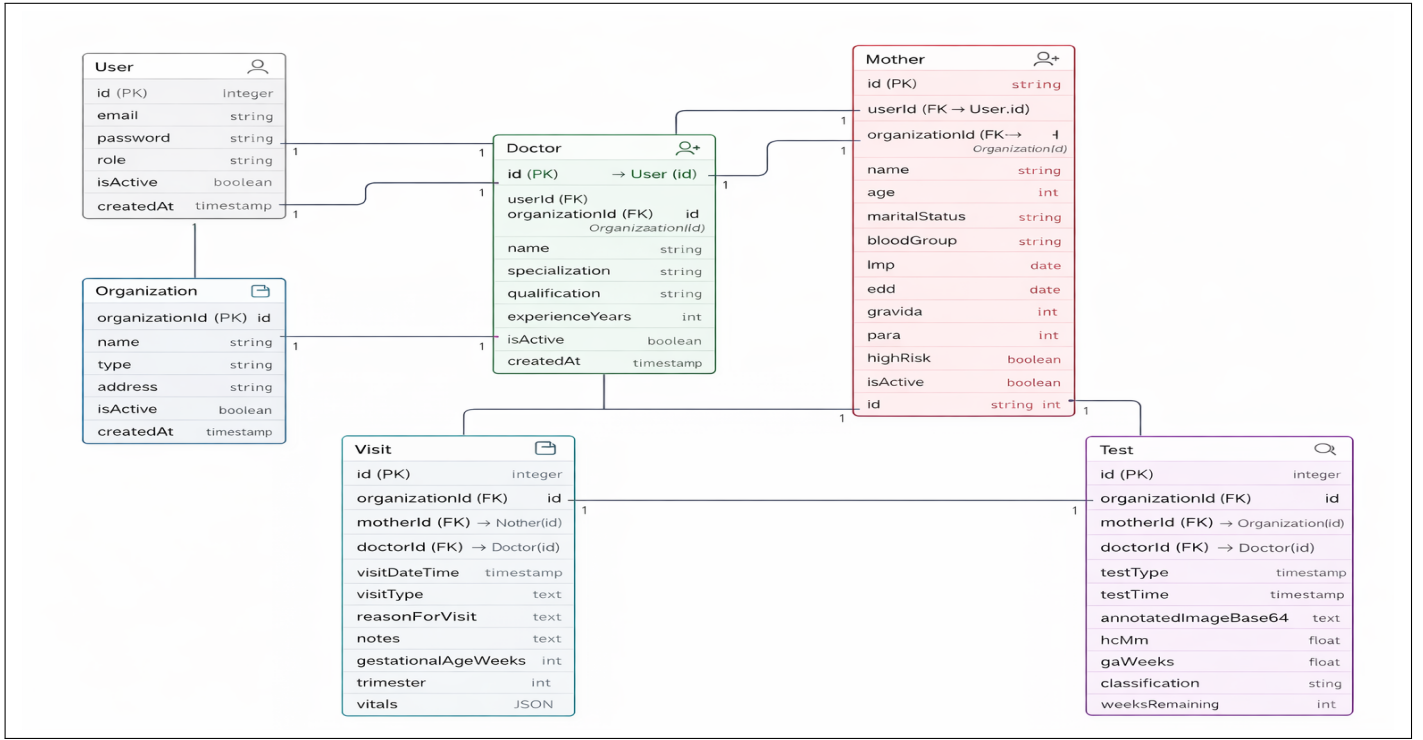


Fig. 5: Entity Relationship Diagram

C. Frontend Implementation

The frontend of Natalis is developed using Flutter, enabling a unified and cross-platform codebase for seamless deployment across multiple devices. The client-side application communicates exclusively with the backend through secure RESTful APIs, ensuring controlled and authenticated data exchange. The frontend workflow follows a structured navigation model that begins with user authentication and session initialization, after which the user interface is dynamically rendered based on the authenticated role, such as doctor, organization, or administrator. In the primary operational flow, a doctor logs into the system and initiates the creation of a new diagnostic test by entering maternal details and uploading the corresponding medical scan. If the maternal profile does not already exist, a new mother record is created and securely stored in the system, following which the diagnostic test is associated with the newly created maternal entity. The system also enables doctors to retrieve and review previously conducted tests, allowing longitudinal access to patient diagnostic history. Data retrieval and visualization are performed through asynchronous API calls, while secure form submission mechanisms ensure the integrity of sensitive medical records. State management is employed to maintain consistency across application screens, and asynchronous request handling ensures a responsive and smooth user experience even during data-intensive operations.

D. Core Workflows and Algorithms

Several core workflows govern the functionality of the Natalis platform, ensuring secure, consistent, and reliable

system operation. The authentication workflow validates user credentials and generates secure authorization tokens, which are subsequently attached to all client requests to enforce authenticated access across services. Upon successful authentication, role-based access control mechanisms dynamically regulate API access, ensuring that system functionalities are exposed only to authorized roles such as doctors, organizations, or administrators. Maternal record management is handled through validated service pipelines that support the creation, update, and retrieval of maternal profiles, with each profile uniquely associated with a registered user and protected by role-based permissions. In the primary operational workflow, a doctor initiates the medical test lifecycle by entering maternal details and uploading the corresponding diagnostic scan. If a maternal profile does not already exist, the system creates a new maternal record and associates it with the initiated test. The Test Service then manages the complete lifecycle of the diagnostic process, including test creation, result storage, and retrieval of historical test records, enabling longitudinal access to patient diagnostics. All workflows are designed to minimize redundancy, enforce validation at multiple layers, and maintain data integrity across service boundaries, thereby ensuring secure and efficient system operation.

E. Tools and Frameworks Used

The implementation of the Natalis platform leverages modern, industry-standard technologies to ensure robustness, scalability, and ease of maintenance. The backend is developed using the Spring Boot framework, incorporating Spring Web

for building RESTful services and Spring Data MongoDB for seamless interaction with the database layer. MongoDB is employed as the primary database to support flexible and scalable storage of healthcare data. The frontend is implemented using Flutter, enabling cross-platform deployment through a unified codebase. Communication between the client and server is conducted via RESTful APIs using JSON-based data exchange, ensuring interoperability and platform independence. Secure access to system resources is enforced through token-based authentication mechanisms. During development and testing, tools such as Postman are used for API validation, Git is employed for version control, and integrated development environment-based debugging tools are utilized to streamline development and ensure code reliability.

V. RESULT AND ANALYSIS

A. Performance Metrics

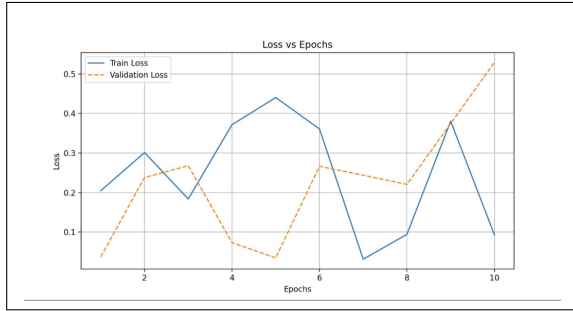


Fig. 6: Loss vs Epochs Graph

The training and validation curves exhibit significant volatility, with the model reaching its peak generalization at epoch 5 before validation loss begins to climb. The sharp divergence observed after epoch 8 confirms that the model is overfitting, as it continues to minimize training error while losing accuracy on unseen data. Implementing early stopping or adjusting the learning rate would likely stabilize these oscillations and improve overall reliability.

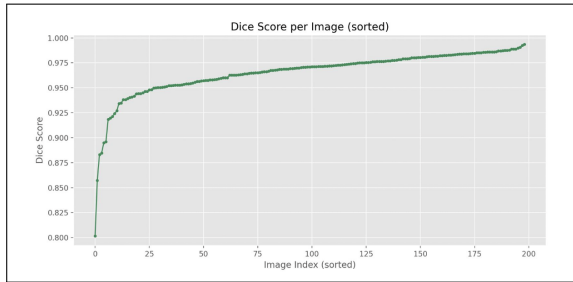


Fig. 7: Dice Score Per Image

The Dice Score per Image plot highlights the model's high level of precision across a dataset of 200 samples. The curve shows that only a negligible fraction of images fall below a 0.90 score, while the vast majority (roughly 85 percent) stabilize in the 0.95 to 0.99 range. This consistency

indicates that the segmentation algorithm is remarkably robust, successfully handling the majority of edge cases with high spatial overlap. Such a high concentration of scores near 1.00 confirms that the model's predictions are nearly identical to the ground truth masks.

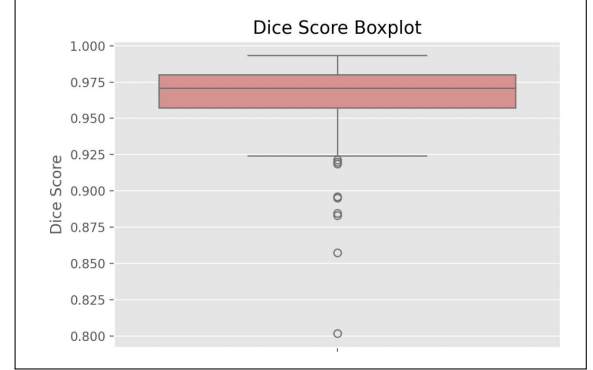


Fig. 8: Dice Score Box Plot

The boxplot reveals a high median score of approximately 0.97, with the majority of data tightly packed between 0.96 and 0.975. This narrow interquartile range demonstrates excellent model consistency and high precision across the test set. While the performance is generally superior, several outliers below 0.92 indicate a few specific images where the segmentation was less effective.

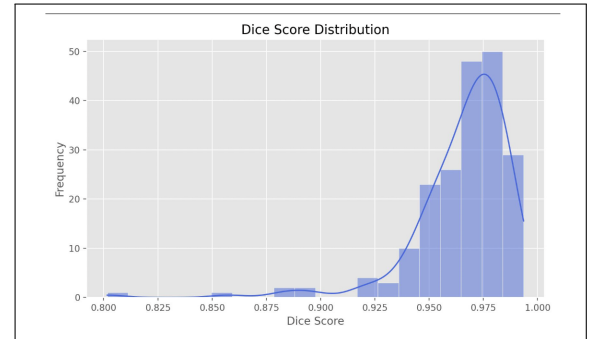


Fig. 9: Dice Score Distribution

The distribution is heavily left-skewed, showing that the vast majority of images achieve a near-perfect score between 0.975 and 1.00. This high frequency in the upper decile confirms the model's robust ability to accurately segment most samples. Very few instances fall below the 0.90 threshold, illustrating that significant errors are rare occurrences for this architecture.

B. Screenshots

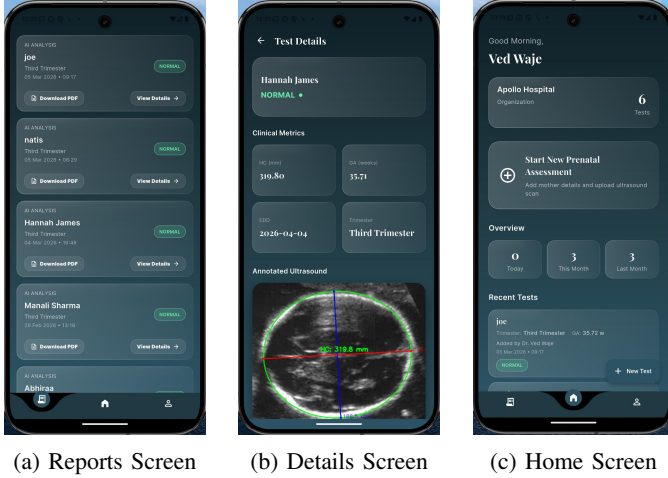


Fig. 10: Screenshots of Application

VI. CONCLUSION

Natalis aims to democratize prenatal screening by bringing AI-powered ultrasound capabilities to underserved and low-resource areas. By working entirely from existing ultrasound images and videos, and running completely offline on a smartphone, Natalis removes common barriers such as unreliable electricity, limited internet connectivity, and the shortage of highly trained sonographers. The system's AI core will provide essential obstetric insights, including gestational age estimation, fetal position, and placental health assessment, directly at the point of care. By guiding health workers through simple steps and delivering clear, actionable results, Natalis will enable the timely identification of high-risk pregnancies that require referral to advanced care. If successfully implemented, Natalis has the potential to significantly improve antenatal care coverage in areas where access to ultrasound is currently limited. More women will be able to receive the WHO-recommended mid-pregnancy scan, and preventable complications could be detected earlier, improving outcomes for both mothers and babies. This project plan sets out a clear pathway from data preparation and AI algorithm development to the deployment of a fully functional mobile application, laying the foundation for a cost-effective, scalable, and life-saving maternal health solution.

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Natalis

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Submitted to Sardar Patel College of Engineering

Student Paper

1%

2

Geocey Shejy, Bhavika Chattani, Varnit Batheja, Manasvi Patwa, Sahil Deshmukh. "Activity Monitoring and Unusual Activity Detection for Elderly Homes", 2023 International Conference on Advanced Computing Technologies and Applications (ICACTA), 2023

Publication

1%

Exclude quotes On

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Exclude matches < 1%

Sustainable Goal: 3 (Good Health & Well-being)

Project Evaluation Sheet 2025 - 26

Title of Project:

NATALIS - Personal Image Interpretation System

Group Members:

Ved Vija D17A / Abhishek Mox D17A /
Aditya D17A / Pooja Bannor D17A /
Harshita Lohara D17A /

Engineering Concepts & Knowledge	Interpretation of Problem & Analysis	Design / Prototype	Interpretation of Data & Dataset	Modern Tool Usage	Societal Benefit, Safety Consideration	Environment Friendly	Ethics	Team work	Presentati on Skills	Applied Engg&M gmt principles	Life - long learning	Profess ional Skills	Innov ative Appr oach	Resear- ch Paper	Total Marks
(5)	(5)	(5)	(3)	(5)	(2)	(2)	(2)	(2)	(2)	(3)	(3)	(3)	(3)	(5)	(50)
5	5	5	3	5	2	2	2	2	2	3	3	3	3	3	48

Comments: Local Cashed data → Remote ecosystem can be planned

Name & Signature Reviewer 1

Engineering Concepts & Knowledge	Interpretation of Problem & Analysis	Design / Prototype	Interpretation of Data & Dataset	Modern Tool Usage	Societal Benefit, Safety Consideration	Environment Friendly	Ethics	Team work	Presentati on Skills	Applied Engg&M gmt principles	Life - long learning	Profess ional Skills	Innov ative Appr oach	Resear- ch Paper	Total Marks
(5)	(5)	(5)	(3)	(5)	(2)	(2)	(2)	(2)	(2)	(3)	(3)	(3)	(3)	(5)	(50)
5	5	5	3	5	2	2	2	2	2	3	3	3	3	3	48

Comments:

Self Application created

Date: 7th February, 2026

Name & Signature Reviewer 2

Vidya . S . 2026

Inhouse/ Industry CareNX Innovation

Innovation/Research:

Project Evaluation Sheet 2025 - 26

Class: D1 / A/B/C

Group No.: 23

Title of Project: Natalis: Prenatal Imaging Interpretation System

Group Members: Ved Waje Abhirat More Pranika Bannore Harshita Lohana

Engineering Concepts & Knowledge	Interpretation of Problem & Analysis	Design / Prototype	Interpretation of Data & Dataset	Modern Tool Usage	Societal Benefit, Safety Consideration	Environment Friendly	Ethics	Team work	Presentation Skills	Applied Engg&Mgmt principles	Life - long learning	Professional Skills	Innovative Approach	Research Paper	Total Marks
(5)	(5)	(5)	(3)	(5)	(2)	(2)	(2)	(2)	(2)	(3)	(3)	(3)	(3)	(5)	(50)
5	4	5	2	5	2	2	2	2	2	3	3	3	3	3	46

Comments: Data Pre-processing for diff. image formats / resolution / noise need to be addressed

Pallavi S.
Name & Signature Reviewer 1

Engineering Concepts & Knowledge	Interpretation of Problem & Analysis	Design / Prototype	Interpretation of Data & Dataset	Modern Tool Usage	Societal Benefit, Safety Consideration	Environment Friendly	Ethics	Team work	Presentation Skills	Applied Engg&Mgmt principles	Life - long learning	Professional Skills	Innovative Approach	Research Paper	Total Marks
(5)	(5)	(5)	(3)	(5)	(2)	(2)	(2)	(2)	(2)	(3)	(3)	(3)	(3)	(5)	(50)
5	4	5	2	5	2	2	2	2	2	3	3	3	3	3	46

Comments: Report description in user friendly manner

Date: 5th March, 2026

Vidya S. Zope
Name & Signature Reviewer 2